Table 3.6 Theoretical and Design Moment Amplification Factors for a Fixed-Ended Beam-Column Loaded by a Uniformly Distributed Lateral Load

$u=\dfrac{kL}{2}=\dfrac{\pi}{2}\sqrt{\dfrac{P}{P_{ek}}}$	Theoretical Eq. (3.6.31)	Design Eq. (3.6.19)
0	0	0
0.20	1.003	1.002
0.40	1.011	1.010
0.60	1.025	1.023
0.80	1.045	1.042
1.00	1.074	1.068
1.20	1.111	1.102
1.40	1.161	1.149
$\pi/2$	∞	∞

form

$$y=-\frac{(M_A\cos kL+M_B)}{EIk^2\sin kL}\sin kx+\frac{M_A}{EIk^2}\cos kx+\frac{M_A+M_B}{LEIk^2}x-\frac{M_A}{EIk^2} \tag{3.7.1}$$

Rearranging, we have

$$y=-\frac{1}{EIk^2}\left[\frac{\cos kL}{\sin kL}\sin kx-\cos kx-\frac{x}{L}+1\right]M_A-\frac{1}{EIk^2}\left[\frac{1}{\sin kL}\sin kx-\frac{x}{L}\right]M_B \tag{3.7.2}$$

from which

Equations (3.7.4) and (3.7.5) can be written in matrix form as

$$\begin{bmatrix}\theta_A\\ \theta_B\end{bmatrix}=\begin{bmatrix}f_{11} & f_{12}\\ f_{21} & f_{22}\end{bmatrix}\begin{bmatrix}M_A\\ M_B\end{bmatrix} \tag{3.7.6}$$

where

$$f_{11}=f_{22}=\frac{L}{EI}\left[\frac{\sin kL-kL\cos kL}{(kL)^2\sin kL}\right] \tag{3.7.7}$$

$$f_{12}=f_{21}=\frac{L}{EI}\left[\frac{\sin kL-kL}{(kL)^2\sin kL}\right] \tag{3.7.8}$$

From Eq. (3.7.6), we can write

$$\begin{bmatrix}M_A\\ M_B\end{bmatrix}=\begin{bmatrix}f_{11} & f_{12}\\ f_{21} & f_{22}\end{bmatrix}^{-1}\begin{bmatrix}\theta_A\\ \theta_B\end{bmatrix} \tag{3.7.9}$$

Table 3.7 Stability Functions ($kL=\pi\sqrt{P/P_e}$)

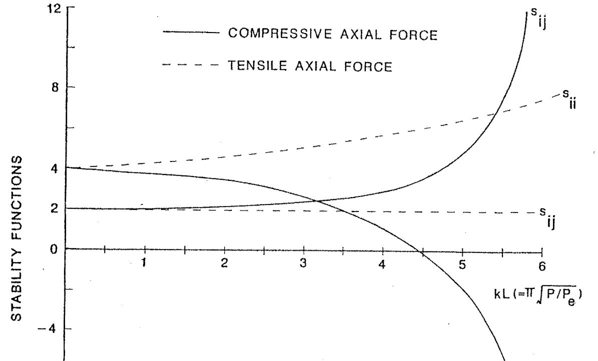
kL	P/P_e	compression s_{ii}	compression s_{ij}	tension s_{ii}	tension s_{ij}
0.	0.	4.0000	2.0000	4.0000	2.0000
0.0500	0.0003	3.9997	2.0001	4.0003	1.9999
0.1000	0.0010	3.9987	2.0003	4.0013	1.9997
0.1500	0.0023	3.9970	2.0008	4.0030	1.9993
0.2000	0.0041	3.9947	2.0013	4.0053	1.9987
0.2500	0.0063	3.9917	2.0021	4.0083	1.9979
0.3000	0.0091	3.9876	2.0028	4.0120	1.9970
0.3500	0.0124	3.9833	2.0039	4.0157	1.9956
0.4000	0.0162	3.9786	2.0054	4.0211	1.9946
0.4500	0.0205	3.9729	2.0068	4.0268	1.9932
0.5000	0.0253	3.9665	2.0084	4.0332	1.9917
0.5500	0.0306	3.9595	2.0102	4.0401	1.9900
0.6000	0.0365	3.9517	2.0121	4.0477	1.9881

Table 3.7 Stability Functions ($kL = \pi\sqrt{P/P_e}$) (*continued*)

kL	P/P_e	compression s_{ii}	compression s_{ij}	tension s_{ii}	tension s_{ij}
2.3500	0.5595	3.2032	2.2213	4.6886	1.8431
2.4000	0.5836	3.1659	2.2328	4.7163	1.8374
2.4500	0.6082	3.1274	2.2447	4.7444	1.8317
2.5000	0.6333	3.0878	2.2572	4.7730	1.8259
2.5500	0.6588	3.0471	2.2701	4.8020	1.8201
2.6000	0.6849	3.0052	2.2834	4.8314	1.8142
2.6500	0.7115	2.9622	2.2974	4.8612	1.8083
2.7000	0.7386	2.9179	2.3118	4.8915	1.8024
2.7500	0.7662	2.8723	2.3268	4.9221	1.7965
2.8000	0.7944	2.8254	2.3425	4.9531	1.7905
2.8500	0.8230	2.7772	2.3587	4.9845	1.7845
2.9000	0.8521	2.7276	2.3756	5.0162	1.7785
2.9500	0.8817	2.6766	2.3932	5.0484	1.7725
3.0000	0.9119	2.6242	2.4115	5.0809	1.7665

Table 3.7 (*continued*)

kL	P/P_e	compression s_{ii}	compression s_{ij}	tension s_{ii}	tension s_{ij}
4.7000	2.2382	−0.6582	3.9839	6.3572	1.5709
4.7500	2.2861	−0.8387	4.0934	6.3987	1.5658
4.8000	2.3344	−1.0289	4.2112	6.4403	1.5606
4.8500	2.3833	−1.2299	4.3381	6.4821	1.5556
4.9000	2.4327	−1.4427	4.4751	6.5241	1.5505
4.9500	2.4826	−1.6685	4.6235	6.5662	1.5456
5.0000	2.5330	−1.9087	4.7845	6.6085	1.5406
5.0500	2.5839	−2.1651	4.9599	6.6509	1.5357
5.1000	2.6354	−2.4394	5.1514	6.6934	1.5309
5.1500	2.6873	−2.7341	5.3613	6.7362	1.5261
5.2000	2.7397	−3.0516	5.5921	6.7790	1.5213
5.2500	2.7927	−3.3953	5.8470	6.8220	1.5166
5.3000	2.8461	−3.7689	6.1297	6.8652	1.5120
5.3500	2.9001	−4.1770	6.4447	6.9084	1.5074
5.4000	2.9545	−4.6254	6.7977	6.9518	1.5028



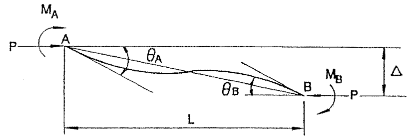
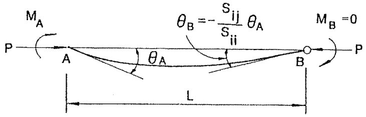


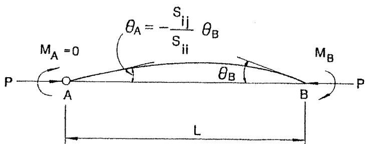
FIGURE 3.19 Beam-column subjected to end moments (with relative joint translation)

as

$$M_A = \frac{EI}{L}\left[s_{ii}\left(\theta_A - \frac{\Delta}{L}\right) + s_{ij}\left(\theta_B - \frac{\Delta}{L}\right)\right]$$



(a)



rotations at joints A and B by θ_A and θ_B, respectively, as in the preceding cases, then the member end rotations, with respect to its chord, will be $(\theta_A - M_A/R_{kA})$ and $(\theta_B - M_B/R_{kB})$. As a result, the modified slope-deflection equations are modified to

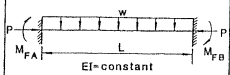
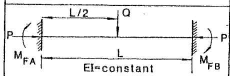
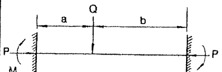
$$M_A = \frac{EI}{L}\left[s_{ii}\left(\theta_A - \frac{M_A}{R_{kA}}\right) + s_{ij}\left(\theta_B - \frac{M_B}{R_{kB}}\right)\right] \tag{3.8.6}$$

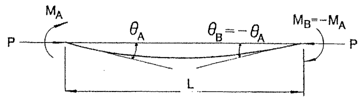
$$M_B = \frac{EI}{L}\left[s_{ij}\left(\theta_A - \frac{M_A}{R_{kA}}\right) + s_{jj}\left(\theta_B - \frac{M_B}{R_{kB}}\right)\right] \tag{3.8.7}$$

Solving Eqs. (3.8.6) and (3.8.7) simultaneously for M_A and M_B gives

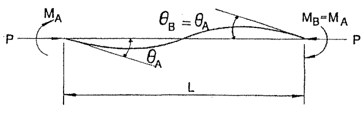
$$M_A = \frac{EI}{LR^*}\left[\left(s_{ii} + \frac{EIs_{ii}^2}{LR_{kB}} - \frac{EIs_{ij}^2}{LR_{kB}}\right)\theta_A + s_{ij}\theta_B\right] \tag{3.8.8}$$

Table 3.8 Expressions for Fixed-End Moments $\left(u = \frac{L}{2}\sqrt{\frac{P}{EI}}\right)$

Case	Fixed-End Moments
	$M_{FA} = -M_{FB} = -\frac{wL^2}{12}\left[\frac{3(\tan u - u)}{u^2 \tan u}\right]$
	$M_{FA} = -M_{FB} = -\frac{QL}{8}\left[\frac{2(1-\cos u)}{u \sin u}\right]$
	$M_{FA} = -\frac{QL}{d}\left[\frac{2ub}{L}\cos 2u - 2u\cos\frac{2ub}{L} - \sin 2u + \sin\frac{2ua}{L} + \sin\frac{2ub}{L} + \frac{2ua}{L}\right]$



(a) SINGLE CURVATURE



made, approximate solutions to some specific cases of inelastic beam-columns can be obtained analytically.[10,11]

In this section, the behavior and failure load of an eccentrically loaded beam-column of rectangular cross section will be investigated. The derivation follows closely to that given in reference 11.

The three basic assumptions used in the following derivation are the following:

1. The deflected shape of the member follows a half-sine wave (Fig. 3.23a).
2. The equilibrium condition is established only at midspan of the member.
3. The stress–strain relationship is assumed to be elastic–perfectly plastic (Fig. 3.23b).

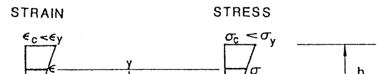
In the process of analysis, we will need to use the nonlinear relationship between the internal moment (M) and the curvature

($\Phi = -y''$) with the presence of an axial force (P). It is therefore necessary to develop this relationship before proceeding to the analysis.

3.9.1 M–Φ–P Relationship

Figure 3.24, a–c, shows a series of strain and stress diagrams that correspond to three stages of loading sequences: the elastic (no yielding), the primary plastic (yielding in compression zone only), and the secondary plastic (yielding in both compression and tension zones),

FIGURE 3.24 Strain and stress diagrams



respectively. For convenience, they are designated in the following as Case 1, Case 2, and Case 3, respectively. The M–Φ–P relationship for each case will be developed separately as follows:

Case 1: Elastic (Fig. 3.24a)

KINEMATICS

From the kinematic assumption that plane sections remain plane after bending, we write

$$\varepsilon = \varepsilon_0 + \Phi y \quad \text{for} \quad -\frac{h}{2} \leqslant y \leqslant \frac{h}{2} \tag{3.9.1}$$

where ε_0 is the axial strain at the centroid of the cross section.

STRESS–STRAIN RELATION

Since the entire cross section is elastic, the stress is related to the strain by

Equation (3.9.6) is the familiar elastic beam moment-curvature relationship of $M = -EIy''$.

Introducing the notations

$$P_y = A\sigma_y = bh\sigma_y \tag{3.9.7}$$

$$M_y = S\sigma_y = \frac{bh^2}{6}\sigma_y \tag{3.9.8}$$

$$\Phi_y = \frac{2\varepsilon_y}{h} = \frac{2\sigma_y}{Eh} \tag{3.9.9}$$

$$p = \frac{P}{P_y} \tag{3.9.10}$$

$$m = \frac{M}{M_y} \tag{3.9.11}$$

STRESS–STRAIN RELATION

$$\sigma = E\varepsilon \quad \text{for} \quad -\frac{h}{2} \leqslant y \leqslant \bar{h} \tag{3.9.18}$$

$$\sigma = \sigma_y \quad \text{for} \quad \bar{h} \leqslant y \leqslant \frac{h}{2} \tag{3.9.19}$$

EQUILIBRIUM

$$P = \int_A \sigma \, dA$$
$$= \int_{-h/2}^{\bar{h}} E[\varepsilon_y - (\bar{h} - y)\Phi] b \, dy + \int_{\bar{h}}^{h/2} \sigma_y b \, dy \tag{3.9.20}$$

$$M = \int \sigma y \, dA$$

STRESS–STRAIN RELATION

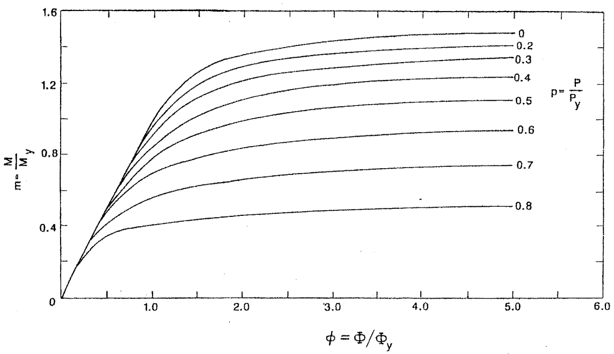
$$\sigma = -\sigma_y \quad \text{for} \quad -\frac{h}{2} \leqslant y \leqslant -\bar{g} \tag{3.9.24}$$

$$\sigma = E\varepsilon \quad \text{for} \quad -\bar{g} \leqslant y \leqslant \bar{h} \tag{3.9.25}$$

$$\sigma = \sigma_y \quad \text{for} \quad \bar{h} \leqslant y \leqslant \frac{h}{2} \tag{3.9.26}$$

EQUILIBRIUM

$$P = \int_A \sigma \, da$$
$$= \int_{-h/2}^{-\bar{g}} (-\sigma_y) b \, dy + \int_{-\bar{g}}^{\bar{h}} E[\varepsilon_y - (\bar{h} - y)\Phi] b \, dy$$



loaded beam-column (Fig. 3.23a) by using the analysis method presented in reference 11. By assuming that the deflected shape of the member resembles a half-sine wave, we write

$$y = \delta \sin \frac{\pi x}{L} \tag{3.9.31}$$

from which we obtain

$$y' = \delta \frac{\pi}{L} \cos \frac{\pi x}{L} \tag{3.9.32}$$

and

$$y'' = -\delta \left(\frac{\pi}{L}\right)^2 \sin \frac{\pi x}{L} \tag{3.9.33}$$

The curvature at midspan is

Equation (3.9.35) can be written as

$$m_0 + \frac{P\delta}{M_y} = m_m \tag{3.9.41}$$

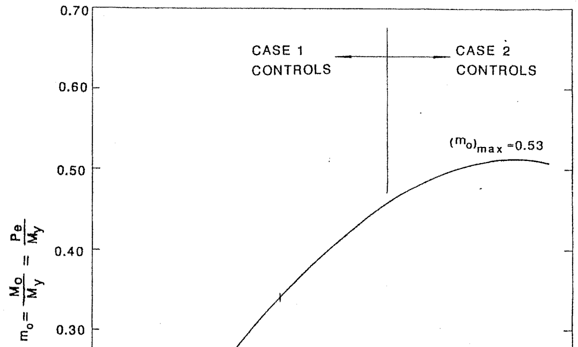
Since, from Eq. (3.9.34)

$$\frac{P\delta}{M_y} = \frac{P\left[\Phi_m\left(\frac{L}{\pi}\right)^2\right]}{EI\Phi_y} = \left(\frac{P}{P_e}\right)\left(\frac{\Phi_m}{\Phi_y}\right) \tag{3.9.42}$$

We can write Eq. (3.9.41) as

$$m_0 + \frac{p}{p_e}\phi_m = m_m \tag{3.9.43}$$

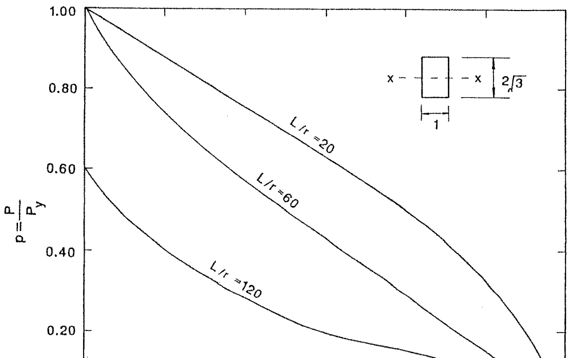
If we substitute the moment-curvature-thrust ($m - \phi - p$) relationships



the failure load may occur in the secondary plastic range [i.e., $\phi_m > 1/(1-p)$]. In this example, the maximum end moment $(m_0)_{max}$ has been obtained graphically as the peak point of the m_0 versus ϕ_m curve. The same maximum moment can be obtained more conveniently by realizing that the peak point of the curve corresponds to the condition

$$\frac{dm_0}{d\phi_m} = 0 \tag{3.9.45}$$

Thus, by setting the derivative of Eq. (3.9.44b, c) with respect to ϕ_m equal to zero, the value of ϕ_m that corresponds to $(m_0)_{max}$ for each relevant case can be calculated. Backsubstituting the value of ϕ_m so obtained into the corresponding equation will give the value of $(m_0)_{max}$. In doing so, a relation between $(m_0)_{max}$ and p can be established as the following:



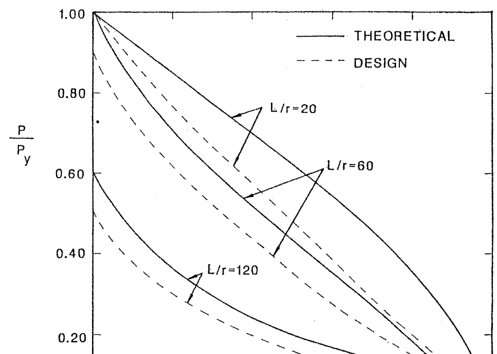
linear interaction equation of the form

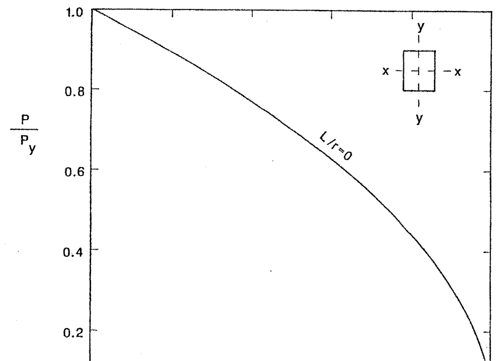
$$\frac{P}{P_u}+\frac{M_{max}}{M_u}=1 \tag{3.10.1}$$

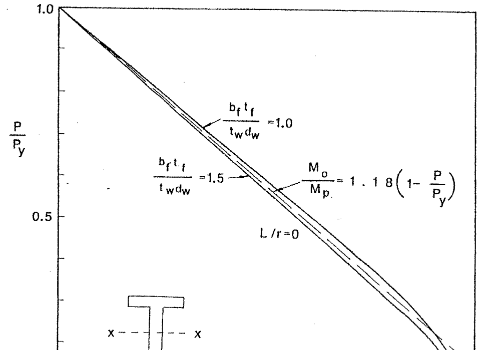
where

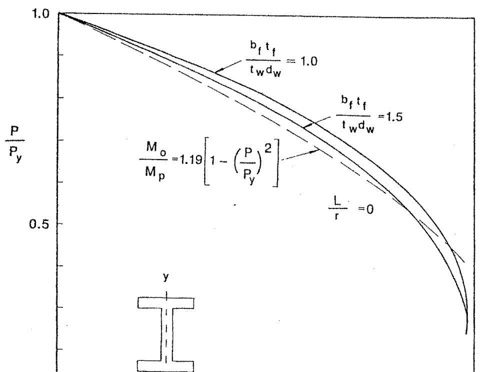
P = axial force in the member
P_u = ultimate axial capacity of the member in the absence of primary bending moment
M_{max} = maximum moment in the member
M_u = ultimate moment capacity of the member in the absence of axial force

Depending on the design philosophy, the ultimate axial capacity of the member P_u can be represented by the CRC curve and SSRC curves or the LRFD curve (Chapter 2). The quantity M_{max}, however, is the




1.0
0.8
0.6
0.4
0.2
P/Py
L/r=0
x
x
y
y


1.0
0.5
P/Py
bf tf / tw dw =1.0
bf tf / tw dw =1.5
Mo/Mp = 1.18(1- P/Py)
L/r=0
x
x



3.10.1 AISC/ASD Format

The AISC/ASD format is a stress-based design format. As a result, allowable stresses and service loads rather than ultimate strengths and maximum loads are used as the basis for design. Equations (3.10.7) and (3.10.8) can be converted to unit of stresses and a factor of safety applies to them to bring them into the service load range.

If stability controls, the interaction equation is

$$\frac{f_a}{F_a}+\frac{f_{bx}C_{mx}}{F_{bx}(1-f_a/F'_{ex})}+\frac{f_{by}C_{my}}{F_{by}(1-f_a/F'_{ey})}\leq 1.0 \qquad (3.10.9)$$

where

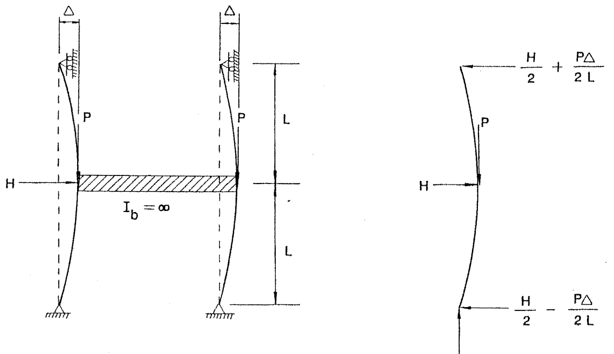
$f_a = P/A_g$ = axial stress at service load

f_{bx}, f_{by} = flexural stresses at service load due to primary bending moment about the x and y axes, respectively

F_a = allowable compressive stress if the member is under axial

Table 3.9 Values for ψ and C_m

Case	ψ	C_m
1	0	1.0
2	−0.4	$1-0.4\ P/P_{ek}$
3	−0.4	$1-0.4\ P/P_{ek}$
4		



3.10.2 AISC/PD Format

The AISC/PD format provides two interaction equations for the design of beam-columns.

If yielding controls,

$$\frac{P}{P_y}+\frac{M}{1.18M_p}\leqslant 1.0 \tag{3.10.12}$$

where

$P_y = A_g F_y =$ yield load of the section where A_g is the area of the cross section

$M_P = ZF_y =$ full plastic moment capacity of the section where Z is plastic-section modulus

P and M are the factored axial force and moment (service loads time load factor). The load factor is 1.7 if only gravity loads are acting and is 1.3 if

3.10.3 AISC/LRFD Format

The AISC/LRFD format based on the exact inelastic solutions of 82 beam-columns,[13] recommends the following interaction equations for sway and nonsway beam-columns.

For $P/\phi_c P_u \geqslant 0.2$

$$\frac{P}{\phi_c P_u}+\frac{8}{9}\left(\frac{M_{ax}}{\phi_b M_{ux}}+\frac{M_{ay}}{\phi_b M_{uy}}\right)\leqslant 1.0 \tag{3.10.15}$$

For $P/\phi_c P_u < 0.2$

$$\frac{P}{2\phi_c P_u}+\frac{M_{ax}}{\phi_b M_{ux}}+\frac{M_{ay}}{\phi_b M_{uy}}\leqslant 1.0 \tag{3.10.16}$$

where

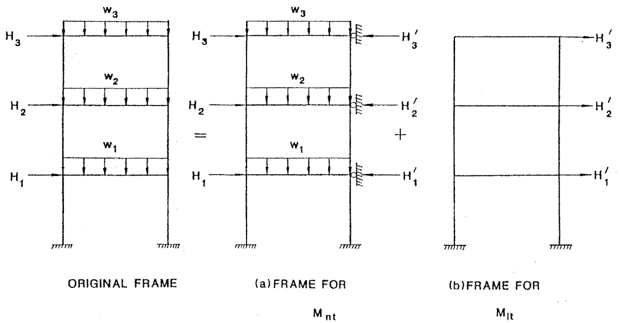


FIGURE 3.34 AISC/LRFD approach for calculating M_{nt} and M_{lt}

(i) each story can behave independently of other stories, and (ii) the additional moments in the columns caused by the $P-\Delta$ effect is equivalent to that caused by a lateral force of $\sum P\Delta/L$, the sway stiffness of the story can be defined as

$$S_F = \frac{\text{horizontal force}}{\text{lateral displacement}}$$

$$= \frac{\sum H}{\Delta_0} = \frac{\sum H + \sum P\Delta/L}{\Delta} \tag{3.10.21}$$

Solving Eq. (3.10.21) for Δ gives

$$\Delta = \left(\frac{1}{1 - \dfrac{\sum P\Delta_0}{\sum HL}} \right) \Delta_0 \tag{3.10.22}$$

If rigid connections and elastic behavior are assumed, the moment amplified moment resulting from the $P-\delta$ effect (i.e., the term B_1M_{nt}) and the amplified moment resulting from the $P-\Delta$ effect (i.e., the term B_2M_{lt}) do not necessarily coincide at the same location. For elastic behavior, the $P-\Delta$ effect usually magnifies the end moments. Nevertheless, because of the assumptions involved in developing the $P-\delta$ and $P-\Delta$ amplification factors, as well as the difficulties involved in locating the exact location of each of the magnified moments in the member, Eq. (3.10.17) gives a justifiable estimation of the design moment for the member.

Note that, unlike the ASD and PD interaction equations in which both the yielding and stability interactions equations are needed in the design process, only one interaction equation is needed if the LRFD approach is used. The applicable equation is determined by the term P/ϕ_cP_u. If $P/\phi_cP_u \geqslant 0.2$, Eq. (3.10.15) is applicable, and if $P/\phi_cP_u < 0.2$, Eq. (3.10.16) is applicable. Another feature of the LRFD approach that is different from the ASD and PD approaches is that the $P-\delta$ and $P-\Delta$

$$M'_{px} = 1.2M_{px}[1-(P/P_y)] \leqslant M_{px} \tag{3.10.30}$$

$$M'_{py} = 1.2M_{py}[1-(P/P_y)^2] \leqslant M_{py} \tag{3.10.31}$$

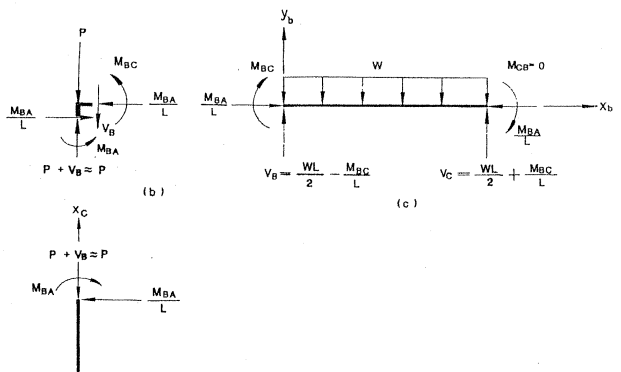
$$M'_{nx} = M_{ux}[1-(P/\phi_cP_u)][1-(P/P_{ex})] \tag{3.10.32}$$

$$M'_{ny} = M_{uy}[1-(P/\phi_cP_u)][1-(P/P_{ey})] \tag{3.10.33}$$

The nonlinear interaction equations expressed in Eqs. (3.10.26) and (3.10.27) were developed by Tebedge and Chen[19] based on curve-fitting to theoretical elastic-plastic beam-column solutions.[3]

3.11 AN ILLUSTRATIVE EXAMPLE

Our discussion of the behavior of beam-columns in this chapter focuses primarily on isolated members. In reality, most structural members exist as parts of a framework and their behavior is therefore influenced by the



Upon rearranging and using the notation $k^2 = P/EI$, Eq. (3.11.1) becomes

$$y_c'' + k^2 y_c = -\frac{M_{BA}}{LEI} x_c \tag{3.11.2}$$

The general solution of Eq. (3.11.2) is

$$y_c = A \sin kx_c + B \cos kx_c - \frac{M_{BA}}{LEIk^2} x_c \tag{3.11.3}$$

The constants A and B can be evaluated by using the boundary conditions

$$y_c(0) = 0 \tag{3.11.4}$$

$$y_c(L) = 0 \tag{3.11.5}$$

It can easily be shown that by using Eqs. (3.11.4) and (3.11.5)

The lowest value of kx_c satisfying the above equation is

$$kx_c = \frac{\pi}{2} \tag{3.11.14}$$

Substituting Eq. (3.11.14) into the moment equation (3.11.10) gives the value of the in-span maximum moment as

$$(M_c)_{max} = \frac{M_{BA}}{\sin kL} \tag{3.11.15}$$

In Eq. (3.11.15), M_{BA} is the column end moment at B whose value can be expressed as a function of the applied load w and P by consideration of joint equilibrium and compatibility at B as demonstrated in the following.

Beam Analysis

by setting the shear in the beam equal to zero:

$$V_b = \left(\frac{wL}{2} - \frac{M_{BC}}{L}\right) - wx_b = 0 \tag{3.11.21}$$

which gives

$$x_b = \frac{L}{2} - \frac{M_{BC}}{wL} \tag{3.11.22}$$

The value of the in-span maximum moment in the beam is then obtained by evaluating the moment at the distance x_b given by Eq. (3.11.22)

$$(M_b)_{max} = M\,\Big|_{x_b = L/2 - M_{BC}/wL}$$

$$= \left[M_{BC} + \left(\frac{wl}{2} - \frac{M_{BC}}{L}\right)x_b - \frac{wx_b^2}{2} \right]$$

In summary, the maximum in-span moment in the column is given by

$$(M_c)_{max} = \frac{M_{BA}}{\sin kL} \tag{3.11.28}$$

and the maximum in-span moment in the beam is given by

$$(M_b)_{max} = \frac{M_{BC}^2}{2wL^2} + \frac{M_{BC}}{2} + \frac{wL^2}{8} \tag{3.11.29}$$

where M_{BA} and M_{BC} are given by Eq. (3.11.27).

If the applied column force P is zero, it can easily be shown by using the L'Hospital rule in Eq. (3.11.27) or by a direct first-order analysis that the column and beam-end moments are given by

$$M_{BA} = -M_{BC} = \frac{wL^2}{16} \tag{3.11.30}$$

Figure 3.37 shows a plot of the maximum-beam moment $(M_{max})_{beam}$, the maximum-column moment $(M_{max})_{column}$, and the beam-end moment M_{BC} $(= -M_{BA}$, the column-end moment) nondimensionalized by the quantity $wL^2/8$ as a function of the applied column force P nondimensionalized by the Euler load. The maximum beam moment is obtained as the larger value of the beam-end moment M_{BC} and the maximum in-span beam moment $(M_b)_{max}$. Similarly, the maximum-column moment is obtained as the larger value of the column-end moment M_{BA} and the maximum in-span column moment $(M_c)_{max}$. The sign convention used in the figure is that a positive-beam moment will cause tension on the bottom fiber of the beam and a positive-column moment will cause tension on the right-side fiber of the column.

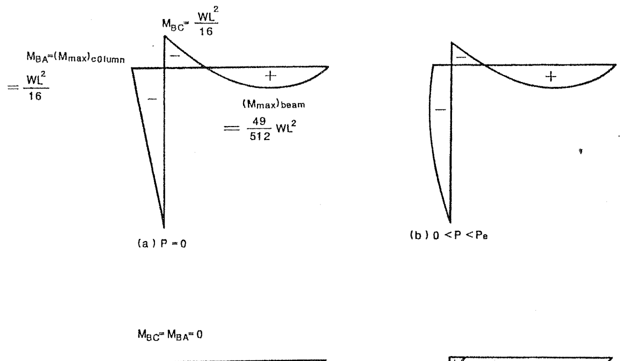
Also shown in the figure are the maximum moments in the column as predicted by the AISC approach [Eq. (3.4.27)], that is $(M_{AB} = 0)$

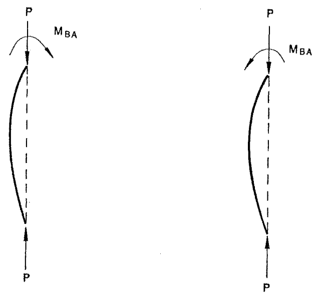
The location of the maximum in-span beam moment is given by

$$x_b = \frac{L}{2} - \frac{M_{BC}}{wL} = \frac{L}{2} + \frac{L}{8}\left[\frac{\pi^2 P/P_e}{3 - 3\pi\sqrt{P/P_e}\cot(\pi\sqrt{P/P_e}) + (\pi^2 P/P_e)}\right] \tag{3.11.34}$$

In writing the above equation, the expression for M_{BC} given in Eq. (3.11.27) with $kL = \pi\sqrt{P/P_e}$ is used. It should be noted that the above expressions for x_c and x_b are valid only if the calculated value falls within the range 0 to L. If the calculated values fall outside this specific range, the location of the maximum moment is at the end rather than within the span of the member. Figure 3.38 shows a plot of the variation of x_c and x_b nondimensionalized by the length of the member L as a function of P/P_e. For the column, the location of the maximum moment shifts from the upper end to the middle of the

FIGURE 3.38 Variation of the location of the maximum





(a) $P < P_e$ (b) $P > P_e$

FIGURE 3.40 Beam-column interaction

the form

$$y^{\mathrm{IV}} + k^2 y'' = \frac{w(x)}{EI} \tag{3.12.1}$$

where $k^2 = P/EI$. The general solution to Eq. (3.12.1) is

$$y = A \sin kx + B \cos kx + Cx + D + f(x) \tag{3.12.2}$$

If there are regions of constant transverse shear force V in the member, if may be more convenient to write the differential equation as

$$y''' + k^2 y' = -\frac{V}{EI} \tag{3.12.3}$$

whose general solution is

$$y = A \sin kx + B \cos kx + C + f(x) \tag{3.12.4}$$

Alternatively, one can draw a free-body diagram of a segment of

where

M_0 = maximum moment that would exist if the axial force in the member were absent (also referred to as the first-order moment)

$A_F = P - \delta$ moment amplification factor to reflect the effect of axial force on magnifying the primary moment in the member

C_m = defined as follows:

1. For members subjected to transverse loadings

$$C_m = 1 + \psi P/P_{ek} \tag{3.12.8}$$

2. For members subjected to end moments only without transverse loadings

$$C_m = 0.6 - 0.4(M_A/M_B) \geqslant 0.4 \tag{3.12.9}$$

The limiting condition $C_m \geqslant 0.4$ has been removed in the LRFD Specification.

because, in many cases, a larger percentage of the nonsway moment is not affected by the $P - \Delta$ effect.

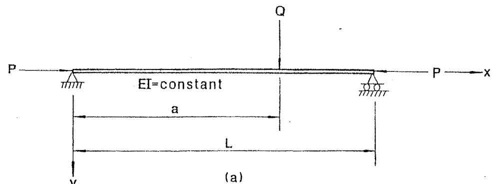
The design of the beam-columns is facilitated by the use of interaction equations. These are equations that relate a combination of axial force and moments that will initiate the failure of a beam-column. They generally give good approximations to the more exact interaction curves developed on the basis of an inelastic analysis. Since inelastic beam-column analysis is rather complicated, interaction equations provide an attractive alternative for designers. The design of beam-columns in various interaction formats as provided by the current AISC Specifications is discussed in Sec. 3.10.

PROBLEMS

3.1 Use the design amplification factor for the lateral deflection y_{max} in Eq. (3.2.35) to derive the design moment amplification factor for the moment M_{max} in Eq. (3.2.41).

is used to approximate the fixed-end moment where $P_{ek} = \pi^2 EI/(KL)^2$ and M_0 = fixed-end moment that would exist in the member if P were absent (first-order moment).

3.5 Using the deflection function given in Eq. (3.3.13a, b) for the beam-column shown in Fig. P3.5a, formulate the expression for the deflection y for the beam-columns shown in Fig. P3.5b-d by the principle of superposition.



(a)

3.6 For the beam-column subjected to end-moments M_A and M_B as shown in Fig. P3.6, find the elastic maximum moment for $P/P_e = 0.4$ if

a. $M_A/M_B = 0.4$
b. $M_A/M_B = 0$
c. $M_A/M_B = -0.4$

Where is the location of M_{max}?

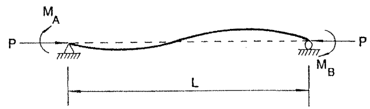
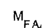


FIGURE P3.6

3.7 Find the fixed-end forces M_{FA} and M_{FB} for the beam-column shown in Fig. P3.7.



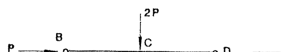
versus P/P_e, where P is the axial force in the column and P_e is the critical load of the column for column DE of the frame shown in Fig. P3.8. What observation do you draw?

3.10 Find the fixed-ended moments M_{FA} and M_{FB} for the unsymmetrically loaded beam-column shown as Case 3 in Table 3.8.

3.11 Determine the exact C_m factor for the beam-column shown in Fig. 3.33(b).

3.12 Using the slope deflection Eqs. (3.8.1) and (3.8.2), determine the critical load of the frame shown in Fig. 3.33(a).

3.13 Find the design moments for the structure shown in Fig. P.3.13 using the LRFD method.



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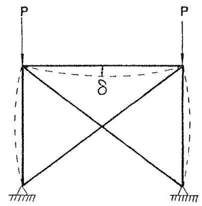
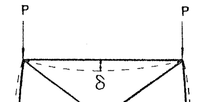
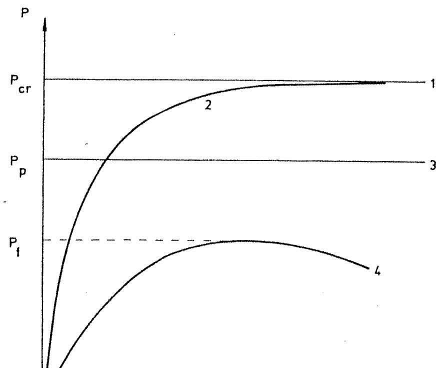
Chapter 4

RIGID FRAMES



(a) PERFECT SYSTEM



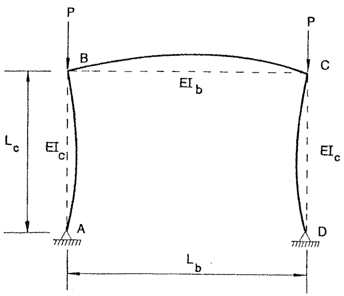


eigenvalue of the system of equations generated by enforcing equilibrium and compatibility conditions of all members and joints in the structural system.

Toward the end of this chapter, we will give a simple method making use of the virtual work principle for the determination of the rigid plastic collapse load P_p. This will be followed by the presentation of a simple interaction equation making use of P_{cr} and P_p to estimate the true failure load P_f of the frame.

To conclude the chapter, we will discuss *the effective length factor K* as recommended by the AISC Specification for the design of members in a framed structure (references 7 and 8 in Chapter 3).

4.2 ELASTIC CRITICAL LOADS BY DIFFERENTIAL EQUATION METHOD



(a)

from which

$$y_c' = \frac{M_B}{P}\left(\frac{1}{L_c} - \frac{k_c \cos k_c x_c}{\sin k_c L_c}\right) \tag{4.2.8}$$

and

$$y_c'(L_c) = \frac{M_B}{P}\left(\frac{1}{L_c} - \frac{k_c}{\tan k_c L_c}\right) \tag{4.2.9}$$

For the beam, the effect of axial force on the behavior of the member is usually negligible (Fig. 4.3b). Therefore, the differential equation has the simple form

$$EI_b y_b'' = M_B \tag{4.2.10}$$

or

$$y_b'' = \frac{M_B}{EI_b} \tag{4.2.11}$$

equation (4.2.19), we have

$$\frac{M_B}{P}\left(\frac{1}{L_c}-\frac{k_c}{\tan k_cL_c}\right)=\frac{-M_BL_b}{2EI_b} \tag{4.2.20}$$

or, rearranging and denoting $k_b^2 = P/EI_b$,

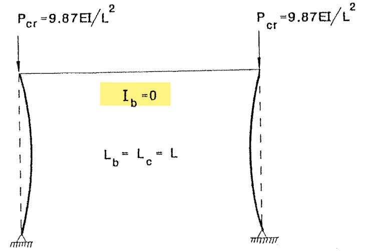
$$\left(\frac{k_b^2L_b}{2}+\frac{1}{L_c}-\frac{k_c}{\tan k_cL_c}\right)M_B=0 \tag{4.2.21}$$

At bifurcation, M_B increases without bound. For Eq. (4.2.21) to be valid, the term in the parenthesis must be zero, i.e.,

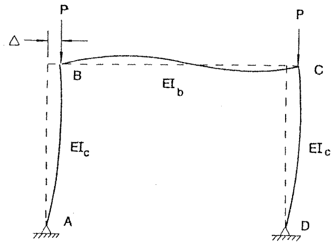
$$\frac{k_b^2L_b}{2}+\frac{1}{L_c}-\frac{k_c}{\tan k_cL_c}=0 \tag{4.2.22}$$

or

$$k_b^2L_bL_c \tan k_cL_c + 2\tan k_cL_c - 2k_cL_c = 0 \tag{4.2.23}$$



(a)



(a)

Therefore, the column's lateral deflection can be expressed in terms of the sway deflection Δ of the frame as

$$y_c = \frac{\Delta}{\sin k_c L_c} \sin k_c x_c \tag{4.2.35}$$

From equilibrium consideration of the column, we have

$$M_B = P\Delta \tag{4.2.36}$$

from which we obtain the sway deflection

$$\Delta = \frac{M_B}{P} \tag{4.2.37}$$

Upon substituting Eq. (4.2.37) into Eq. (4.2.35), we obtain the deflection function of the column as

Substituting Eqs. (4.2.45) and (4.2.46) into Eq. (4.2.43) gives the deflection function of the beam as

$$y_b = \frac{M_B L_b}{6EI_b} x_b + \frac{M_B}{EI_b}\left(\frac{x_b^3}{3L_b} - \frac{x_b^2}{2}\right) \tag{4.2.47}$$

from which

$$y_b' = \frac{M_B L_b}{6EI_b} + \frac{M_B}{EI_b}\left(\frac{x_b^2}{L_b} - x_b\right) \tag{4.2.48}$$

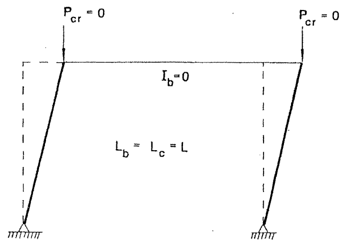
and

$$y_b'(0) = \frac{M_B L_b}{6EI_b} \tag{4.2.49}$$

Joint compatibility at B requires that

$$y_c'(L_c) = y_b'(0) \tag{4.2.50}$$

or, using Eq. (4.2.40) and Eq. (4.2.49), we obtain



the boundary conditions and joint compatibility conditions of each member and each joint from which the charactersitic equation of the frame can be obtained. The eigenvalue of this characteristic equation will give the critical load of the frame.

Although the differential equation method can, in theory, be used to determine P_{cr} for all types of frames, the actual implementation of the method for the solution of a given frame is rather complex, especially when it is applied to frames of more than one story and one bay. Fortunately, there is a simpler method available, which makes use of the slope-deflection equations[1] developed in the previous chapter. This will be discussed in the following section.

4.3 ELASTIC CRITICAL LOADS BY SLOPE-DEFLECTION EQUATION METHOD

In the slope-deflection equation method, the slope-deflection equations developed in Sections 3.7 and 3.8 are written for each and every member

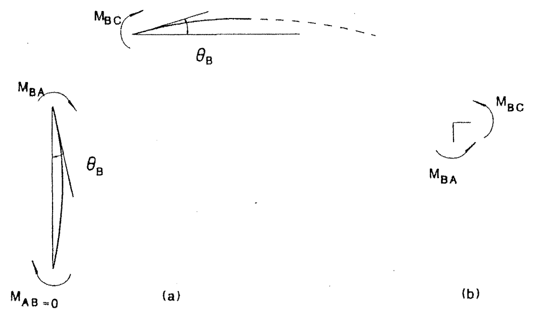


FIGURE 4.7 Slope-deflection equation approach for P_{cr} of nonsway buckling of simple portal frame.

or

$$s_{iic} - \frac{s_{ijc}^2}{s_{iic}} + 2 = 0 \tag{4.3.9}$$

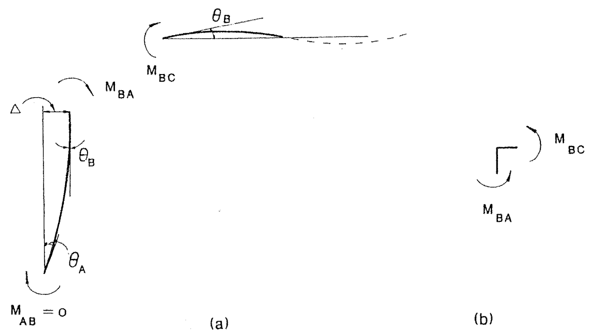
By trial and error and using Table 3.7, the value of kL that satisfies Eq. (4.3.9) is found to be

$$kL = (\sqrt{P/EI})L = 3.59 \tag{4.3.10}$$

from which the critical load

$$P_{cr} = 12.9\frac{EI}{L^2} \tag{4.3.11}$$

is obtained. This load is the same as before using the differential equation approach.



P P

From story shear equilibrium (Fig. 4.8c), we have

$$\frac{M_{AB}+M_{BA}+P\Delta}{L_c}+\frac{M_{CD}+M_{DC}+P\Delta}{L_c}=0 \tag{4.3.20}$$

Realizing that

$$M_{AB}=M_{DC}=0 \quad \text{(hinged)} \tag{4.3.21}$$

and

$$M_{CD}=M_{BA} \quad \text{(antisymmetry)} \tag{4.3.22}$$

we can write the story-shear equilibrium equation (4.3.20) as

$$\frac{2M_{BA}+2P\Delta}{L_c}=0 \tag{4.3.23}$$

or

$$\frac{M_{BA}+P\Delta}{L_c}=0 \tag{4.3.24}$$

(4.3.28) to be valid, we must set

$$\det\begin{vmatrix} S+6 & -S \\ -S & S-(kL)^2 \end{vmatrix}=0 \tag{4.3.29}$$

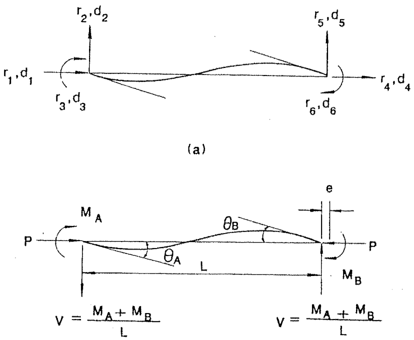
Equation (4.3.29) is the characteristic equation of the frame. By trial and error and by using Table 3.7, the value kL can be found to be

$$kL=(\sqrt{P/EI})L=1.35 \tag{4.3.30}$$

from which the critical load can be solved

$$P_{cr}=1.82\frac{EI}{L^2} \tag{4.3.31}$$

Note the correspondence of Eq. (4.3.31) obtained using the slope-deflection method with Eq. (4.2.57) obtained previously using the differential equation method.



Equations (4.4.1) to (4.4.6) can be written in matrix form as

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\dfrac{1}{L} & -\dfrac{1}{L} \\ 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & \dfrac{1}{L} & \dfrac{1}{L} \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} P \\ M_A \\ M_B \end{pmatrix} \tag{4.4.10}$$

Similarly, Eqs. (4.7) to (4.4.9) can be written in matrix form as

Substituting Eq. (4.4.15) into Eq. (4.4.10), and then substituting Eq. (4.4.11) into the resulting equation, we can relate the element end forces (r_1 to r_6) with the element end displacements (d_1 to d_6) as

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \end{pmatrix}_{\text{ns}} = \frac{EI}{L} \begin{bmatrix} \frac{A}{I} & 0 & 0 & -\frac{A}{I} & 0 & 0 \\ & \frac{2(s_{ii}+s_{ij})}{L^2} & -\frac{(s_{ii}+s_{ij})}{L} & 0 & \frac{-2(s_{ii}+s_{ij})}{L^2} & \frac{-(s_{ii}+s_{ij})}{L} \\ & & s_{ii} & 0 & \frac{s_{ii}+s_{ij}}{L} & s_{ij} \\ & \text{sym.} & & \frac{A}{I} & 0 & 0 \\ & & & & \frac{2(s_{ii}+s_{ij})}{L^2} & \frac{(s_{ii}+s_{ij})}{L} \\ & & & & & s_{ii} \end{bmatrix}_{\text{ns}} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{pmatrix}$$

where the subscript s is used to indicate the quantities due to sidesway of the member.

By combining Eq. (4.4.17) and Eq. (4.4.20), we obtain the general beam-column element force-displacement relationship as

$$\mathbf{r} = \mathbf{k}\mathbf{d} \tag{4.4.21}$$

where

$$\mathbf{r} = \mathbf{r}_{\text{ns}} + \mathbf{r}_{\text{s}} \tag{4.4.22a}$$

$$\mathbf{k} = \mathbf{k}_{\text{ns}} + \mathbf{k}_{\text{s}} \tag{4.4.22b}$$

$$\mathbf{k} = \frac{EI}{L} \begin{bmatrix} \frac{A}{I} & 0 & 0 & -\frac{A}{I} & 0 & 0 \\ & \frac{2(s_{ii}+s_{ij})-(kL)^2}{L^2} & -\frac{(s_{ii}+s_{ij})}{L} & 0 & \frac{-2(s_{ii}+s_{ij})+(kL)^2}{L^2} & \frac{-(s_{ii}+s_{ij})}{L} \\ & & & 0 & s_{ii}+s_{ij} & \end{bmatrix}$$

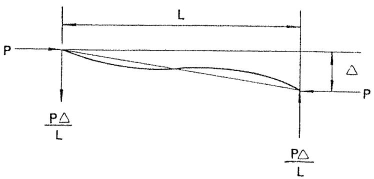


FIGURE 4.10 Additional shear due to swaying of the member

indefinite. However, by using the L'Hospital's rule, it can be shown that these functions will approach unity and Eq. (4.4.24) reduces to the first-order (linear) element stiffness matrix for a frame member.

Also shown in Table 4.1 are the ϕ_i functions expressed in the form of a

Table 4.1 Expressions for ϕ_1, ϕ_2, ϕ_3, and ϕ_4

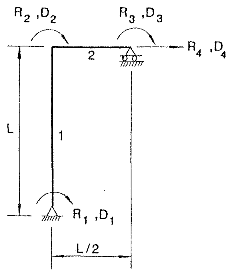
ϕ	P		
	Compressive	Zero	Tensile
ϕ_1	$\dfrac{(kL)^3 \sin kL}{12\phi_c}$	1	$\dfrac{(kL)^3 \sinh kL}{12\phi_t}$
ϕ_2	$\dfrac{(kL)^2(1-\cos kL)}{6\phi_c}$	1	$\dfrac{(kL)^2(\cosh kL - 1)}{6\phi_t}$
ϕ_3	$\dfrac{(kL)(\sin kL - kL\cos kL)}{4\phi_c}$	1	$\dfrac{(kL)(kL\cosh kL - \sinh kL)}{4\phi_t}$
ϕ_4	$\dfrac{(kL)(kL-\sin kL)}{2\phi_c}$	1	$\dfrac{(kL)(\sinh kL - kL)}{2\phi_t}$

where

$$\phi_c = 2 - 2\cos kL - kL\sin kL \qquad \phi_t = 2 - 2\cosh kL + kL\sinh kL$$

matrix that is valid for small axial force is given by

$$
\mathbf{k} = \frac{EI}{L}\begin{bmatrix}
\frac{A}{I} & 0 & 0 & -\frac{A}{I} & 0 & 0 \\
 & \frac{12}{L^2} & \frac{-6}{L} & 0 & \frac{-12}{L^2} & \frac{-6}{L} \\
 & & 4 & 0 & \frac{6}{L} & 2 \\
 & & & \frac{A}{I} & 0 & 0 \\
 & & & & \frac{12}{L^2} & \frac{6}{L} \\
 & & & & & 4
\end{bmatrix}
\mp \frac{P}{L}\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 \\
 & \frac{6}{5} & \frac{-L}{10} & 0 & \frac{-6}{5} & \frac{-L}{10} \\
 & & \frac{2L^2}{15} & 0 & \frac{L}{10} & \frac{-L^2}{30} \\
 & & & 0 & 0 & 0 \\
 & & & & \frac{6}{5} & \frac{L}{10} \\
 & & & & & \frac{2L^2}{15}
\end{bmatrix}
$$



(a)

This stiffness matrix relates the four end forces (r_2, r_3, r_5, and r_6) to the four end displacements (d_2, d_3, d_5, and d_6) of an inextensible member. Note that the element stiffness matrix for an inextensible member [Eq. (4.4.29)] is obtained simply by deleting the first and fourth rows and the first and fourth columns from the element stiffness matrix for an extensible member [Eq. (4.4.27)].

Figure 4.11b shows the four degrees of freedom (d_2, d_3, d_5, and d_6) and the corresponding end forces (r_2, r_3, r_5, and r_6) associated with each member of the structure. Again, the directions are shown in their positive sense in the figure.

By using Eq. (4.4.29), the element stiffness matrix for the column (element 1) can be written as

$$\begin{bmatrix} \frac{12}{L^2} & \frac{-6}{L} & \frac{-12}{L^2} & \frac{-6}{L} \\ \cdots \end{bmatrix} \quad \begin{bmatrix} \frac{6}{5} & \frac{-L}{10} & \frac{-6}{5} & \frac{-L}{10} \\ \cdots \end{bmatrix}$$

relationship is

$$\begin{pmatrix} d_2 \\ d_3 \\ d_5 \\ d_6 \end{pmatrix}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}_1 \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{pmatrix} \tag{4.4.32}$$

For element 2, the kinematic relationship is

$$\begin{pmatrix} d_2 \\ d_3 \\ d_5 \\ d_6 \end{pmatrix}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}_2 \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{pmatrix} \tag{4.4.33}$$

In view of the above observation, Eqs. (4.4.36) and (4.4.37) can be written symbolically as

$$\mathbf{R}_1 = \mathbf{T}_1^{\mathrm{T}} \mathbf{r}_1 \tag{4.4.38}$$

$$\mathbf{R}_2 = \mathbf{T}_2^{\mathrm{T}} \mathbf{r}_2 \tag{4.4.39}$$

From consideration of joint equilibrium, we can write

$$\mathbf{R} = \mathbf{R}_1 + \mathbf{R}_2 \tag{4.4.40}$$

Substituting the member equilibrium relationships Eqs. (4.4.38) and (4.4.39) into Eq. (4.4.40) gives

$$\mathbf{R} = \mathbf{T}_1^{\mathrm{T}} \mathbf{r}_1 + \mathbf{T}_2^{\mathrm{T}} \mathbf{r}_2 \tag{4.4.41}$$

Since, from Eq. (4.4.21) the element force-displacement relationship for elements 1 and 2 can be written, respectively, as

can be written as

$$\mathbf{K} = \frac{EI}{L}\begin{bmatrix} 4 & 2 & 0 & \dfrac{-6}{L} \\ & 12 & 4 & \dfrac{-6}{L} \\ & & 8 & 0 \\ \text{sym.} & & & \dfrac{12}{L^2} \end{bmatrix} - \frac{P}{L}\begin{bmatrix} \dfrac{2L^2}{15} & \dfrac{-L^2}{30} & 0 & \dfrac{-L}{10} \\ & \dfrac{2L^2}{15} & 0 & \dfrac{-L}{10} \\ & & 0 & 0 \\ \text{sym.} & & & \dfrac{6}{5} \end{bmatrix} \tag{4.4.49}$$

Denoting

$$\lambda = \frac{PL^2}{30EI} = \frac{(kL)^2}{30} \tag{4.4.50}$$

Eq. (4.4.49) can be written as

slope-deflection equation approaches. However, it should be noted that the steps shown above can easily be programmed in a digital computer, and so P_{cr} can be obtained quite conveniently for any type of frame.

4.5 SECOND-ORDER ELASTIC ANALYSIS

In the preceding sections, we determined the load that corresponds to a state of bifurcation of equilibrium of a perfect frame by an eigenvalue analysis. In an eigenvalue analysis, the system is assumed to be perfect. There will be no lateral deflections in the members until the load reaches the critical load P_{cr}. At the critical load P_{cr}, the original configuration of the frame ceases to be stable and with a slight disturbance, the lateral deflections of the members begin to increase without bound as indicated by curve 1 in Fig. 4.2. However, if the system is not perfect, lateral deflections will occur as soon as the load is applied, as shown by curve 2 in Fig. 4.2. For an elastic frame, curve 2 will approach curve 1 asymptotically. To trace this curve, a complete load-deflection analysis of

summarized in the following steps (in the following discussion, a subscript refers to the load step and a superscript refers to the cycle of calculation within each load step):

1. First, discretize the frame into a number of beam-column elements.
2. Next, formulate the element stiffness matrix **k** for each and every element. The element stiffness matrix is given in Eq. (4.4.24), or in its approximate form, Eq. (4.4.27). (P can be set equal to zero in these equations for the first cycle of calculations.)
3. Assemble all these element stiffness matrices to form the structure stiffness matrix **K**.
4. Solve for the incremental displacement vector using

$$\Delta \mathbf{R}_i = \mathbf{K}_i^1 \, \Delta \mathbf{D}_i^1 \tag{4.5.1}$$

from which

$$\Delta \mathbf{D}_i^1 = (\mathbf{K}_i^1)^{-1} \, \Delta \mathbf{R}_i \tag{4.5.2}$$

11. Calculate the external force vector from

$$\mathbf{R}_{i+1} = \mathbf{R}_i + \Delta\mathbf{R}_i \tag{4.5.4}$$

12. Evaluate the unbalanced force $\Delta\mathbf{Q}_i^1$ at the end of the cycle from

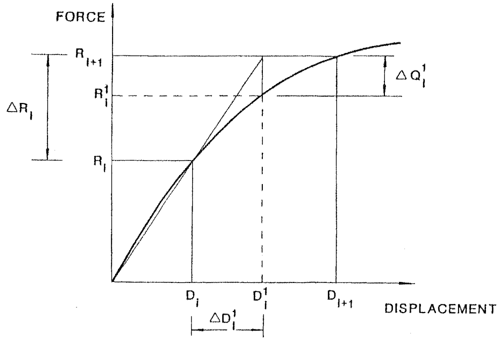
$$\Delta\mathbf{Q}_i^1 = \mathbf{R}_{i+1} - \mathbf{R}_i^1 \tag{4.5.5}$$

13. Using the current value of axial force P, update the element stiffness matrix $\mathbf{k}$ for each and every element. Assemble $\mathbf{k}$ for all the elements to form an updated secant structure stiffness matrix $\mathbf{K}_i^2$. Evaluate the incremental displacement vector $\Delta\mathbf{D}_i^2$ from

$$\Delta\mathbf{D}_i^2 = (\mathbf{K}_i^2)^{-1}\,\Delta\mathbf{Q}_i^1 \tag{4.5.6}$$

where $\Delta\mathbf{Q}_i^1$ is the unbalanced force vector calculated in the previous cycle of calculation.

14. Update the structure nodal displacement vector from



(a) FIRST CYCLE OF CALCULATION

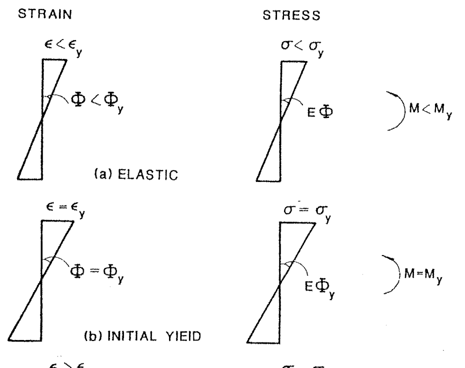
connected rectangular frameworks of usual proportion, the two approaches would give satisfactory results.

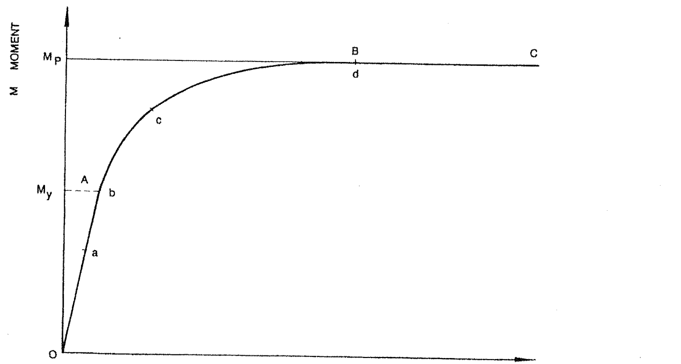
4.6 PLASTIC COLLAPSE LOADS

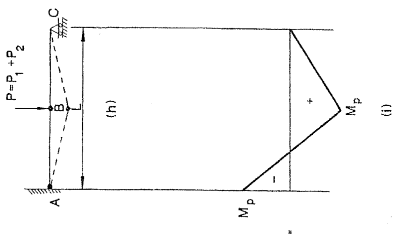
In the preceding discussions, the frame is assumed to behave elastically throughout the entire stage of loading up to failure. The failure of the frame is a result of instability when the stiffness of the frame vanishes. Consequently, the elastic critical load P_{cr} is the maximum load for an *elastic* frame. On the other extreme, if we exclude the instability effect but consider only the plastic yielding of the material, failure of the frame will be controlled by the formation of a *plastic collapse mechanism*. The plastic collapse mechanism load P_p is the maximum load-carrying capacity of the frame. A plastic collapse mechanism will form when there are sufficient number of plastic hinges developed in the structure to render it statically unstable. Before we proceed to the discussion of the

at A marks the *yield point* of the material. The stress level that corresponds to point A is the yield stress of the material. If the stress is below the yield stress σ_y, the material behaves elastically as shown by line OA. After the yield stress has been reached, the strain can increase greatly without any further increase in stress as indicated by line AB. When the strain has reached $\varepsilon_{st} \simeq 12\varepsilon_y$, further increase in strain will bring about a further increase in stress as a result of *strain-hardening*, as indicated by line BC. For simplicity, the effect of strain-hardening is usually not considered in a simple plastic design analysis. Neglecting this effect will obviously lead to a conservative design. When material can sustain a large deformation without fracture, this is known as *ductility*. It is this unique property of structural steel that makes plastic design possible.

When a member is subjected to pure bending, and if the usual assumption of plane sections before bending remain plane after bending


STRAIN
STRESS
ε < ε_y
σ < σ_y
Φ < Φ_y
EΦ
M < M_y
(a) ELASTIC
ε = ε_y
σ = σ_y
Φ = Φ_y
EΦ_y
M = M_y
(b) INITIAL YIEID


M MOMENT
M_P
B
C
d
c
A
M_y
b
a
O



The moment at midspan (point B), when the first hinge is just formed, is

$$\frac{5P_1L}{32}=\frac{5(16M_p/3L)L}{32}=\frac{5}{6}M_p \tag{4.6.3}$$

Figure 4.16d shows the moment diagram at the end of load stage 1.

Load Stage 2

After the formation of the first plastic hinge at A, the propped cantilever becomes a simply supported beam with a constant moment M_p at A to carry the load P_1^*. When the additional load P_2 is applied, the beam behaves as a simply supported member (Fig. 4.16e); Figure 4.16f shows the corresponding moment diagram due to P_2. The load P_2 is now added to the first plastic hinge load P_1^* and the resulting maximum moment at midspan under the combined load $P_1^* + P_2$ is $5M_p/6 + P_2L/4$. The second sequence of formation of plastic hinges is traced. In general, if the structure is statically indeterminate to the nth degree, then the formation of $(n+1)$ plastic hinges will be necessary for the structure to reach its collapse state.

4.6.3 Plastic Collapse Load by Mechanism Method

To determine the plastic collapse P_p in a more direct manner, a simpler method, known as the *mechanism method,* will be presented here. This method is based on the *upper bound theorem* of plastic analysis. The theorem states that a load computed on the basis of an assumed failure or collapse mechanism will always be greater than or at most equal to the true collapse load. Thus, in using this theorem, all possible collapse mechanisms of the structure are identified and the load corresponding to each of these mechanisms is evaluated. The mechanism that gives the lowest value of P_p will be the collapse mechanism. Strictly speaking, to

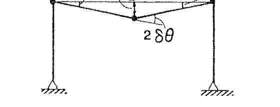
load P_p can be determined directly from the assumed collapse mechanism as shown in Fig. 4.17b. The calculations can be made in the following manner: Assuming the plastic hinge at A undergoes a virtual rotation of $\delta\theta$, it is readily seen from the geometry of the collapse mechanism that the plastic hinge at B will undergo a virtual rotation of $2\delta\theta$, and the vertical drop of point B from its original position is $L\delta\theta/2$. Note that small displacement assumption is used in evaluating this vertical drop.

The external virtual work done during this virtual displacement is equal to the applied load P times the distance it travels, i.e.,

$$\delta W_{\text{ext}} = P(L\delta\theta/2) \tag{4.6.7}$$

and the virtual strain energy stored in the structure during this virtual displacement is equal to the sum of the plastic moment times the hinge rotation at points A and B, i.e.,

$$\delta U_{\text{int}} = M_p(\delta\theta) + M_p(2\delta\theta) \tag{4.6.8}$$



(b) BEAM MECHANISM

Sway Mechanism

$$P(L\delta\theta/2) = M_p(\delta\theta) + M_p(\delta\theta) \tag{4.6.14}$$

from which we obtain

$$P_2 = 4\frac{M_p}{L} \tag{4.6.15}$$

Combined Mechanism

$$(1.5P)(L\delta\theta/2) + P(L\delta\theta/2) = M_p(2\delta\theta) + M_p(2\delta\theta) \tag{4.6.16}$$

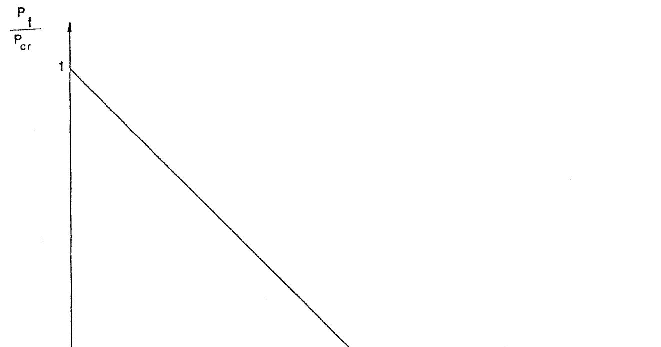
from which we obtain

$$P_3 = \frac{16}{5}\frac{M_p}{L} \tag{4.6.17}$$

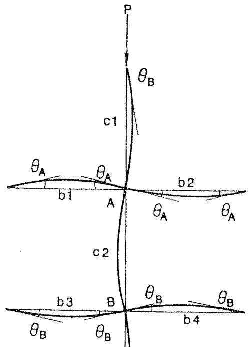
Since the lowest value is P_3, from the upper bound theorem, we therefore choose P_3 as the collapse load. However, to ensure that P_3 is readily seen that the condition $M \leqslant M_p$ is satisfied everywhere in the frame (Fig. 4.19b). Therefore, we can conclude from the lower bound theorem of plastic analysis[10] that $P_p = P_3 = 16M_p/5L$ is the collapse load and the combined mechanism is the collapse mechanism of the frame.

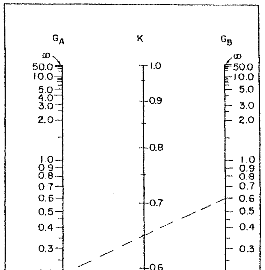
For multistory multibay frames, a number of possible collapse mechanisms will exist. It may be difficult for the analyst to envision all these possible mechanisms. However, various mechanisms can be constructed systematically from the two basic mechanisms, namely the beam and the sway mechanisms, by a process known as *combination of mechanism*. This method is described in detail in reference 10 and will not be presented here. Alternatively, one can obtain P_p for such frames by the hinge-by-hinge method with the aid of a computer.[9]

Generally speaking, the plastic collapse load P_p will give a reasonable estimate of the failure load P_f of the frame only if the effect of instability



taken into account in a design. One convenient way to include this interaction effect is to use the concept of *effective length factor K*. We discussed the concept of effective length factor for isolated column with idealized end conditions thoroughly in Chapter 2. Here we will discuss the effective length factor for a framed member. The determination of the effective length factor K for a framed member is more involved than that for an isolated member, because the stiffness of all adjacent members, as well as the rigidities of the connections, must be included in estimating the rotational restraint at the ends of the member in question. In theory, the effective length factor K for a framed column should be determined from a stability analysis of the entire structure. However, for design purposes this procedure is impractical and the use of a simpler procedure is much more desirable. One such procedure was proposed by Julian and Lawrence.[12] Their procedure was recommended by the AISC, and we will therefore discuss it in what follows. Since the behavior of a framed column will be different, depending on whether





For Beam 4

$$(M_B)_{b4} = \left(\frac{EI}{L}\right)_{b4}[4\theta_B - 2\theta_B] = \left(\frac{EI}{L}\right)_{b4}(2\theta_B) \tag{4.8.8}$$

Note that because of Assumption 2, we can use $s_{ii} = 4$ and $s_{ij} = 2$ for the beams.

For joint equilibrium at A, we must have

$$(M_A)_{c1} + (M_A)_{c2} + (M_A)_{b1} + (M_A)_{b2} = 0 \tag{4.8.9}$$

from which we obtain

$$(M_A)_{c2} = -(M_A)_{b1} - (M_A)_{b2} - (M_A)_{c1} \tag{4.8.10}$$

Substituting Eqs. (4.8.5), (4.8.6), and (4.8.1) for $(M_A)_{b1}$, $(M_A)_{b2}$, and $(M_A)_{c1}$, respectively, into Eq. (4.8.10), we have

where we use the same notation

$$\sum_B \left(\frac{EI}{L}\right)_b = \left(\frac{EI}{L}\right)_{b3} + \left(\frac{EI}{L}\right)_{b4} \tag{4.8.17}$$

$$\sum_B \left(\frac{EI}{L}\right)_c = \left(\frac{EI}{L}\right)_{c2} + \left(\frac{EI}{L}\right)_{c3} \tag{4.8.18}$$

Eliminating $(M_A)_{c2}$ from Eqs. (4.8.2) and (4.8.13), and $(M_B)_{c2}$ from Eqs. (4.8.3) and (4.8.16), we can obtain the following equations

$$\left(s_{ii} + 2\frac{\sum_A \left(\frac{EI}{L}\right)_b}{\sum_A \left(\frac{EI}{L}\right)_c}\right)\theta_A + s_{ij}\theta_B = 0 \tag{4.8.19}$$

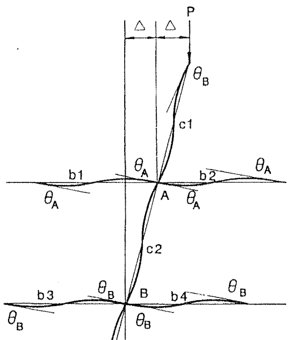
Using the expressions for s_{ii} and s_{ij} [(Eqs. (3.7.15) and (3.7.16)] in Eq. (4.8.24) and realizing that

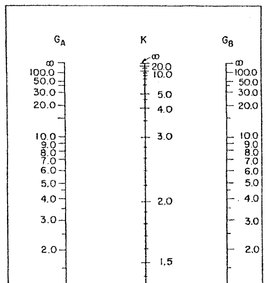
$$kL = (\sqrt{P/EI})L = \pi\sqrt{P/P_e} = \pi/K \tag{4.8.25}$$

Equation (4.8.24) can be simplified to give

$$\frac{G_A G_B}{4}\left(\frac{\pi}{K}\right)^2 + \left(\frac{G_A + G_B}{2}\right)\left(1 - \frac{\pi/K}{\tan(\pi/K)}\right) + \frac{2\tan(\pi/2K)}{\pi/K} - 1 = 0 \tag{4.8.26}$$

Equation (4.8.26) can be expressed in a nomograph form as shown in Fig. 4.21b (see reference 12). To obtain the effective length factor K of column AB, one needs only to evaluate the relative stiffness factors G_A and G_B expressed in Eqs. (4.8.21) and (4.8.22), respectively, at its two ends. A straight line joining the two G values will cut the middle line which gives the value of K. For example, if $G_A = 0.2$, $G_B = 0.6$, the K





from which

$$(M_A)_{c2} = -(M_A)_{b1} - (M_A)_{b2} - (M_A)_{c1} \tag{4.8.36}$$

Substituting Eqs. (4.8.31), (4.8.32), and (4.8.27) for $(M_A)_{b1}$, $(M_A)_{b2}$, and $(M_A)_{c1}$ into Eq. (4.8.36), we have

$$(M_A)_{c2} = -6\left[\left(\frac{EI}{L}\right)_{b1} + \left(\frac{EI}{L}\right)_{b2}\right]\theta_A - \left(\frac{EI}{L}\right)_{c1}\left[s_{ii}\theta_A + s_{ij}\theta_B - (s_{ii}+s_{ij})\frac{\Delta}{L_{c1}}\right] \tag{4.8.37}$$

From Eq. (4.8.28), we can write

$$s_{ii}\theta_A + s_{ij}\theta_B - (s_{ii}+s_{ij})\frac{\Delta}{L_{c2}} = \frac{(M_A)_{c2}}{\left(\frac{EI}{L}\right)_{c2}} \tag{4.8.38}$$

Eliminating $(M_A)_{c2}$ from Eqs. (4.8.28) and (4.8.39), and $(M_B)_{c2}$ from Eqs. (4.8.29) and (4.8.42), we can obtain the following equations

$$\left(s_{ii} + 6\frac{\sum_A \left(\frac{EI}{L}\right)_b}{\sum_A \left(\frac{EI}{L}\right)_c}\right)\theta_A + s_{ij}\theta_B - (s_{ii}+s_{ij})\frac{\Delta}{L_{c2}} = 0 \tag{4.8.45}$$

and

$$s_{ij}\theta_A + \left(s_{ii} + 6\frac{\sum_B \left(\frac{EI}{L}\right)_b}{\sum_B \left(\frac{EI}{L}\right)_c}\right)\theta_B - (s_{ii}+s_{ij})\frac{\Delta}{L_{c2}} = 0 \tag{4.8.46}$$

Equations (4.8.45) and (4.8.46) are obtained by considering joint equilibrium at A and B, respectively. A third equation can be obtained

At bifurcation, we must have

$$\det\begin{vmatrix} s_{ii}+\dfrac{6}{G_A} & s_{ij} & -(s_{ii}+s_{ij}) \\ s_{ij} & s_{ii}+\dfrac{6}{G_B} & -(s_{ii}+s_{ij}) \\ -\dfrac{6}{G_A} & -\dfrac{6}{G_B} & (kL)^2_{c2} \end{vmatrix}=0 \tag{4.8.51}$$

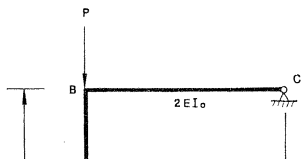
Using the expressions for s_{ii} and s_{ij} [Eqs. (3.7.15) and (3.7.16)] in Eq. (4.8.51) and realizing that $kL=\pi/K$, Eq. (4.8.51) can be simplified to

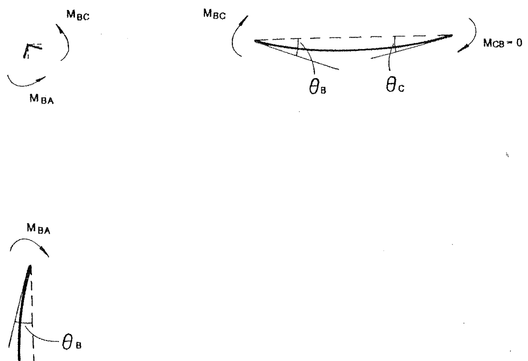
$$\frac{G_A G_B(\pi/K)^2-36}{6(G_A+G_B)}-\frac{(\pi/K)}{\tan(\pi/K)}=0 \tag{4.8.52}$$

4.9 ILLUSTRATIVE EXAMPLES

Example 4.1. Two-Member Frame
For the structure shown in Fig. 4.23, determine the effective length factor K for column AB using

(a) the slope-deflection equation
(b) the nomograph





from which

$$P = P_{cr} = \frac{30.7EI}{L^2} \quad \text{and} \quad K = \sqrt{\frac{P_e}{P_{cr}}} = \sqrt{\frac{\pi^2}{30.7}} = 0.57 \tag{4.9.12}$$

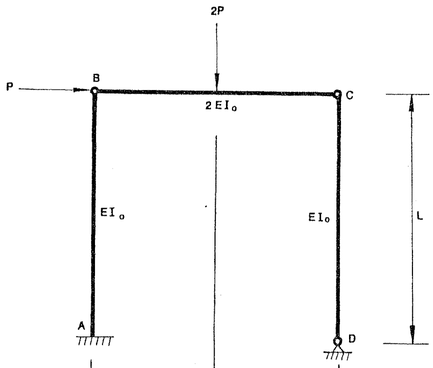
(b) *Nomograph*

$$G_A = 0 \quad \text{(fixed-end)} \tag{4.9.13}$$

$$G_B = \frac{(EI_0/L)}{2EI_0/L} = \frac{1}{2} \tag{4.9.14}$$

The G_B value calculated above is valid only if $\theta_c = -\theta_B$ (see Fig. 4.21). In our case $\theta_C \neq -\theta_B$, and so G_B must be adjusted. The adjustment can be made by realizing that for $\theta_C = -\theta_B$,

$$M_{BC} = \frac{2EI_0}{L}(2\theta_B) = \frac{4EI_0}{L}\theta_B \tag{4.9.15}$$



Column AB

The design moment for column AB is determined from Eq. (3.10.17). The magnitudes for M_{nt} and M_{lt} can easily be determined from Fig. 4.26b as 0 and PL, respectively. To determine the B_1 and B_2 factors, we need to evaluate the effective length K for the column. From the nomograph in Fig. 4.22, K was found to be 2. However, since the right column is pinned at both ends, it cannot resist any sidesway motion. As a consequence, all the resistance to frame instability effect comes from the left column. For this situation, the column that does not provide any sidesway resistance is said to *lean on* the column, which provides the sidesway resistance. Since not all of the columns are effective in resisting sidesway, the effective length of the column providing the sidesway resistance must be modified. Le Messurier[15] suggested a formula for this modification

$$K_i = \sqrt{\frac{\pi^2 EI}{L^2 P_i}\left(\frac{\sum P}{\sum P_{ek}}\right)} \tag{4.9.19}$$

and, from Eq. (3.10.20)

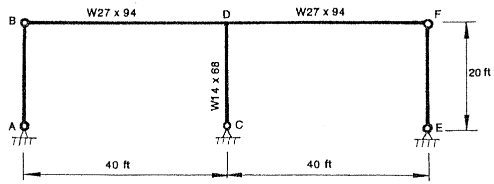
$$B_2 = \frac{1}{1 - \dfrac{\sum P}{\sum P_{ek}}} = \frac{1}{1 - \dfrac{2P}{P_e/(2\sqrt{2})^2}}$$

$$= \frac{1}{1 - 16\dfrac{P}{P_e}} \tag{4.9.22}$$

The design moment is, from Eq. (3.10.17)

$$M_a = B_1 M_{nt} + B_2 M_{lt}$$

$$= \frac{0.6}{1 - 8\dfrac{P}{P_e}}(0) + \frac{1}{1 - 16\dfrac{P}{P_e}}(PL)$$



Section Properties:

W14 x 68

$A = 20.0$ in^2 $I_x = 723$ in^4 $Z_x = 115$ in^3

$r_x = 6.01$ in $r_y = 2.46$ in

W27 x 94

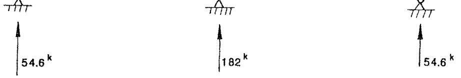


FIGURE 4.28 First-order analysis for load combination 1.2D + 1.6L

Upon comparing λ_{cx} and λ_{cy}, it can be concluded that weak axis bending controls. Using Eq. (2.11.9), P_u was found to be

$$P_u = \exp[-0.419\lambda_{cy}^2]P_y$$
$$= \exp[-0.419(1.09)^2](20.0)(36) = 438 \text{ kips} \tag{4.9.27}$$

Check interaction equation

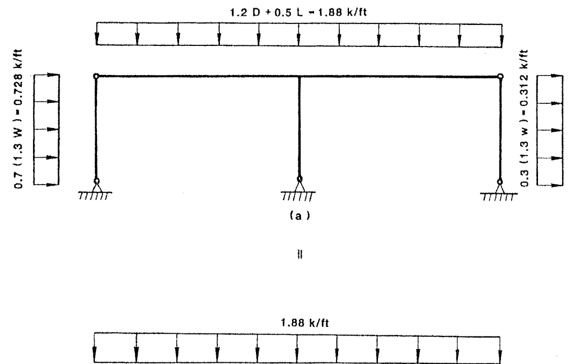
$$\frac{P}{\phi_c P_u} = \frac{182}{(0.85)(438)} = 0.489 > 0.2 \tag{4.9.28}$$

Use the Expression (3.10.15)

$$\frac{P}{\phi_c P_u} + \frac{8}{9}\left(\frac{M_{ax}}{\phi_b M_{ux}} + \frac{M_{ay}}{\phi_b M_{uy}}\right) = 0.489 + 0 + 0 = 0.489 \tag{4.9.29}$$

which is <1; therefore all right.

Load Combination (4): 1.2D + 1.3W + 0.5L



Since the exterior pinned-ended columns lean on column CD, K_x must be adjusted. Using the method proposed in reference 15 as cited in the commentary of LRFD, the modified effective length is given by

$$(K_x)_{\text{adjusted}} = \sqrt{\frac{\pi^2 EI}{L^2 P_{CD}}\left(\frac{\sum P}{\sum P_{ek}}\right)}$$
$$= \sqrt{\frac{\pi^2 EI}{L^2 (94)}\left(\frac{150.4}{P_e/K_x^2 + 0 + 0}\right)}$$
$$= \sqrt{1.6K_x^2} = 2.19 \qquad (4.9.35)$$

and so

$$\lambda_{cx} = \frac{(K_x)_{\text{adjusted}} L}{\pi r_x}\sqrt{\frac{F_y}{E}}$$
$$= \frac{(2.19)(20)(12)}{\quad}\sqrt{\frac{36}{\quad}} = 0.981 \qquad (4.9.36)$$

B_2 factor

$$B_2 = \frac{1}{1 - \dfrac{\sum P}{\sum P_{ek}}}$$

$$= \frac{1}{1 - \dfrac{150.4}{749}} = 1.25 \tag{4.9.40}$$

$$M_{ax} = B_1 M_{nt} + B_2 M_{lt}$$
$$= (1)(0) + 1.25(208)$$
$$= 260 \text{ ft} - \text{kip} = 3120 \text{ in} - \text{kip} \tag{4.9.41}$$

Determine M_{ux}

$$M_{ux} = M_{px} = Z_x F_y = (115)(36) = 4140 \text{ in} - \text{kip} \tag{4.9.42}$$

Check interaction equation

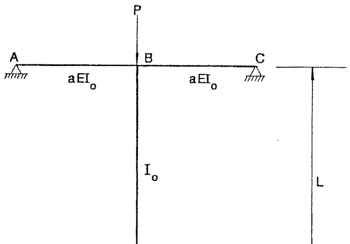
In reality, the collapse loads P_f of most frameworks are neither P_{cr} nor P_p, because collapse in most cases is a result of an interaction of the effects of instability and plasticity. The exact interaction relation is rather complex. However, for design purposes, the Merchant–Rankine interaction equation provides a simple but reasonably accurate method to estimate P_f.

For the design of members in a frame, the K-factor approach provides a convenient means to account for the end restraint effects of other members on the behavior of the member in question. The two nomographs or alignment charts for braced and unbraced frames have been developed to aid designers to obtain K-factors for columns that are component parts of a frame.

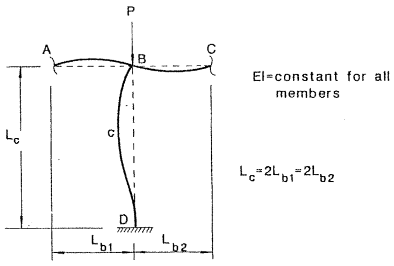
PROBLEMS

4.1 Find P_{cr} for the frame in Fig. P4.1 using

4.3 For the frame in Fig. P4.3,
   a. sketch the buckled shape of the frame
   b. establish upper and lower bounds for the critical load
   c. find the critical load for $a = 1$



4.6 Discuss how the G factors for the nomograph can be adjusted to account for the cases where the far ends of the beams are all
   a. pinned
   b. fixed



11. Horne, M. R., and Merchant, W. The Stability of Frames. Pergamon Press, New York, 1965.

12. Julian, O. G., and Lawrence, L. S. Notes on J and L Nomograms for Determination of Effective Lengths. Unpublished report.

13. Yura, J. A. The effective length of columns in unbraced frames, AISC Engineering Journal, 8(2), April, 1971, pp. 37–42.

14. Wood, R. H. Effective lengths of columns in multistory buildings. Structural Engineers. Vol. 52, Nos. 7, 8, 9, July, August, September, 1974, pp. 235–244, 295–302, 341–346.

15. LeMessuier, W. J. A practical method of second order analysis, part 2—rigid frames, AISC Engineering Journal, 14(2), April, 1977, pp. 49–67.

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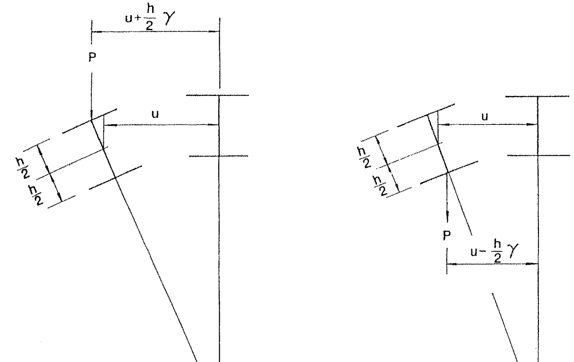
Bleich, F. Buckling Strength of Metal Structures. McGraw-Hill, New York, 1952.

Chajes, A. Principles of Structural Stability Theory. Prentice Hall, Englewood Cliffs, New Jersey, 1974.

Timoshenko, S. P., and Gere, J. M. Theory of Elastic Stability

Chapter 5

BEAMS



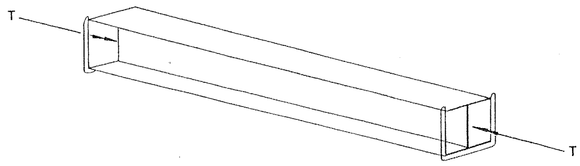


FIGURE 5.2 Simply supported I-beam subjected to twisting moment

5.2 UNIFORM TORSION OF THIN-WALLED OPEN SECTIONS

When an equal and opposite torque T is applied to the ends of a simply supported beam with a thin-walled open section, such as an I-section (Fig. 5.2), the twisting moment along the length of the members is

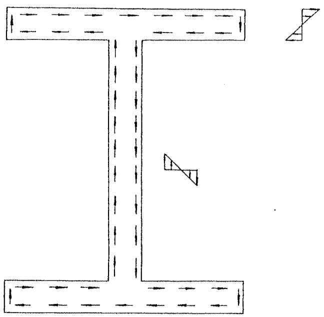
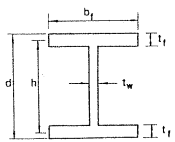


FIGURE 5.4 St. Venant shear stress distribution in an I-section

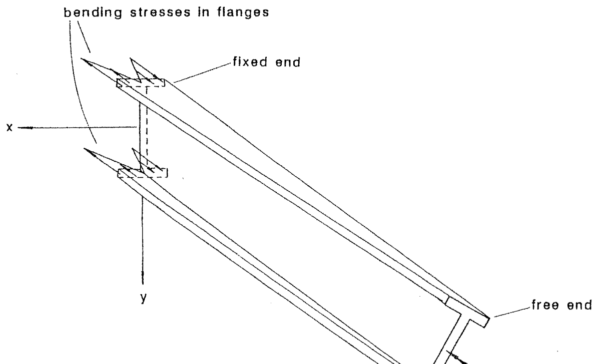
torsion that is associated with these shear stresses is referred to as St.

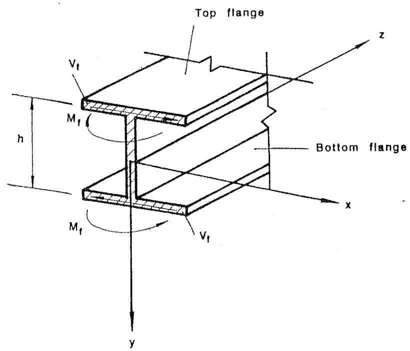
Table 5.1 Torsional Constant and Warping Constant for a Doubly Symmetric I-Section

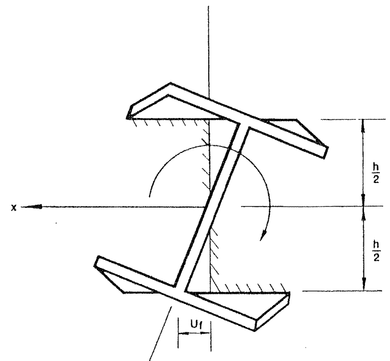
	Torsional Constant	Warping Constant
	$J = \dfrac{2b_f t_f^3 + (d-2t_f)\, t_w^3}{3}$	$C_w = \dfrac{t_f\, b_f^3\, h^2}{24}$

For the purpose of analysis, it is more convenient to express Eq. (5.2.1) in the form of the rate of twist as

$$\frac{d\gamma}{\;} = \frac{T_{sv}}{\;} \tag{5.2.4}$$


bending stresses in flanges
fixed end
x
y
free end


Top flange
z
V_f
M_f
h
Bottom flange
x
M_f
V_f
y



where

$$C_w = \frac{I_f h^2}{2} \tag{5.3.8}$$

is called the *warping constant* of the I-section (Table 5.1). It should be noted that the warping constant is different for different cross sections. A general expression for the warping constant is found in reférence 2; it is

$$C_w = \int_0^s (\bar{w}_s - w_s)^2 t \, ds \tag{5.3.9}$$

in which $w_s = \int_0^s r\,ds$, r equals the distance from the shear center of the cross section to the tangent at any point around the cross section, the equation

$$\bar{w}_s = \frac{1}{s}\int_0^s w_s \, ds$$

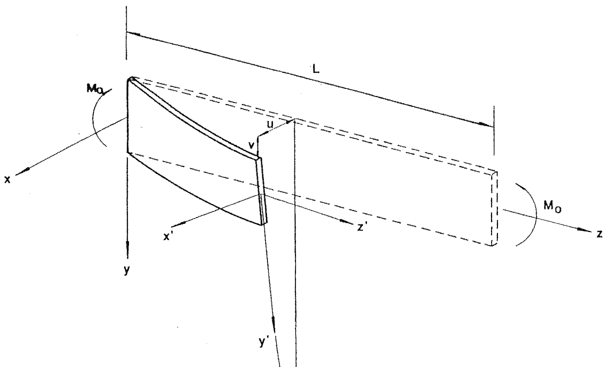
the shear centers 0 for these sections are located at the point of intersection of the component plates. So w_s and $\bar{w}_s$ will be zero. Physically, this means that all these sections do not warp when subjected to twisting moments. Another type of cross sections for which no warping will occur is the axisymmetric sections, such as the solid or circular tubes (see Fig. 5.9b). Here, those sections that are originally plane will remain plane after the twisting moment is applied. For sections other than those shown in Fig. 5.9, warping will generally occur when a twisting moment is applied. Warping for narrow rectangular sections and box sections composed of narrow rectangular elements are usually negligible, and so C_w may be taken as zero for these sections. If warping is restrained, the applied twisting moment will be resisted by both St. Venant torsion and warping restraint torsion.

$$T = T_{sv} + T_w \tag{5.3.10}$$

In view of Eq. (5.2.5) and Eq. (5.3.7), we have

5.4 LATERAL BUCKLING OF BEAMS

When a beam is bent about its axis of greatest flexural rigidity, out-of-plane bending and twisting will occur when the applied load reaches its critical value, unless the beam is provided with a sufficient lateral support. For a geometrically perfect beam, this critical load corresponds to the point of bifurcation of equilibrium when in-plane bending deformation of the member ceases to be stable and out-of-plane bending and twisting deformations become the stable configuration of the member. Here, as in the case of a column, to find the critical load of the beam one must first establish the equilibrium conditions of the beam in a slightly deformed configuration. The critical or lateral buckling load is then obtained as the lowest eigenvalue satisfying the characteristic equation of the differential equations. The following examples will illustrate the procedure for determining this critical load. The assumptions used in the following examples are the following:



chapter, only in-plane bending of the member is considered, whereas here both in-plane and out-of-plane bending, as well as twisting, will be considered. Thus, the lateral torsional buckling problem will be more complex than the in-plane buckling problem, as the former is a three-dimensional problem, whereas the latter is a two-dimensional problem.

For the beam shown in Fig. 5.10, the components of external moments acting on a cross section with a distance z from the origin with respect to the x-y-z coordinates are; $(M_x)_{ext} = M_0$, $(M_y)_{ext} = (M_z)_{ext} = 0$. Figure 5.11a-c shows the moments and moment components in the three mutually perpendicular planes. In this chapter, for convenience, we use a

FIGURE 5.11 Components of moments

right-handed screw rule to represent the moment vector. For example, imagine now a right-hand screw whose axis is colinear with the x axis. If this screw were turned in the direction of M_0 at the positive face $z = L$, as shown in Fig. 5.10, it would tend to advance in the direction of the positive x axis. Because of this, the applied moment M_0 is positive and represented by the positive moment $(M_x)_{ext}$ in Fig. 5.11 in the positive x direction. This sign convention for moment is called the *right-handed screw rule*. Using this sign convention for moment, the moments acting on the cross section in the slightly deflected position with respect to the x'-y'-z' coordinate system can now be obtained directly from Fig. 5.11.

$$(M_{x'})_{ext} \approx (M_x)_{ext} = M_0 \tag{5.4.1}$$

$$(M_{y'})_{ext} \approx -\gamma (M_x)_{ext} = -\gamma M_0 \tag{5.4.2}$$

$$(M_{z'})_{ext} = \frac{du}{dz}(M_x)_{ext} = \frac{du}{dz} M_0 \tag{5.4.3}$$

An inspection of these equations shows that the first equation contains only the variable v and is independent of the other two equations. In fact, this equation describes the in-plane bending behavior of the member that occurs before lateral instability. It is not important for the out-of-plane lateral torsional buckling behavior in a small displacement buckling analysis. The buckling behavior of the beam is described by the last two equations, which are coupled, as they both contain u and γ as variables. If we differentiate Eq. (5.4.9) once with respect to z and substitute the result into Eq. (5.4.8), the two equations can be combined to give

$$\frac{EI_y GJ}{M_0} \frac{d^2\gamma}{dz^2} + \gamma M_0 = 0 \tag{5.4.10}$$

Upon rearranging, and denoting $k^2 = M_0^2/EI_y GJ$, we have the differential equation lateral bending stiffness EI_y and the torsional stiffness GJ. Thus, the coupling effect of out-of-plane deformation and twisting is manifested in the result.

Equation (5.4.16), although derived for a narrow rectangular section, is also valid for box sections composed of narrow rectangular shapes. Like narrow rectangular cross sections, warping for box sections are negligible and so warping constant C_w can be set to zero. However, for box sections, M_{cr} will be much higher due to a significant increase in values of I_y and J.

Example 5.2. Simply Supported I-Section Under Pure Bending
A simply supported I-beam subjected to a pair of equal and opposite end moments applied about the x axis is shown in Fig. 5.12.

SOLUTION: Here, as in the preceding example, two sets of coordinate axes, x-y-z and x'-y'-z', are used to facilitate the analysis. Since no

By equating the corresponding external and internal moments, the governing differential equations for the I-section under pure bending are obtained

$$EI_x \frac{d^2v}{dz^2} + M_0 = 0 \tag{5.4.18}$$

$$EI_y \frac{d^2u}{dz^2} + \gamma M_0 = 0 \tag{5.4.19}$$

$$GJ\frac{d\gamma}{dz} - EC_w \frac{d^3\gamma}{dz^3} - \frac{du}{dz} M_0 = 0 \tag{5.4.20}$$

Again, the first equation is of no interest to us, as it describes the in-plane behavior of the beam before the lateral buckling. The differential equation describing the behavior of the beam at lateral torsion buckling is obtained by combining Eqs. (5.4.19) and (5.4.20) conditions must be satisfied.

$$\left.\frac{d^2\gamma}{dz^2}\right|_{z=0} = 0, \qquad \left.\frac{d^2\gamma}{dz^2}\right|_{z=L} = 0 \tag{5.4.27}$$

From the first conditions of Eqs. (5.4.26) and (5.4.27), we obtain

$$B = 0, \qquad C = -D \tag{5.4.28}$$

and from the second conditions of Eqs. (5.4.26) and (5.4.27), we obtain the simultaneous equations

$$\begin{aligned} A \sin mL - 2D \sinh nL &= 0 \\ Am^2 \sin mL + 2Dn^2 \sinh nL &= 0 \end{aligned} \tag{5.4.29}$$

For a nontrivial solution, the determinant of the above equations must vanish

to Eq. (5.3.8), the warping constant C_w is proportional to the square of the depth h of the section. Thus, *if I_f remains unchanged,* an increase in h will increase C_w and, so, M_{cr}.

In developing Eq. (5.4.34), it has been assumed that the in-plane deflection has no effect on the lateral torsional buckling behavior of the beam. This assumption is justified when the flexural rigidity EI_x is much larger than the flexural rigidity EI_y so that the in-plane deflection will be negligible compared with that of the out-of-plane deflection. If both rigidities are of the same order of magnitude, the effect of bending in the vertical y-z plane may be important and should be considered in calculating M_{cr}. An approximate solution that includes the effect of in-plane deflection is given by Kirby and Nethercot[3] as

$$M_{ocr} = \frac{\pi}{L}\sqrt{\frac{EI_y GJ}{I_r}}\sqrt{(1 + W^2)} \tag{5.4.36}$$

where

In this section, we will present a simple but effective method to take into account the effect of non-uniform moment on the critical lateral buckling loads of beams. Here, as in the beam-column case, the approach is based on the *equivalent moment concept* and the accuracy of the approach has been found to be quite sufficient for most practical cases.

5.5.1 Unequal End Moments

If a beam is subjected to end moments that are unequal in magnitude (Fig. 5.13), the moment in the beam will be a function of z. Consequently, the resulting governing differential equation will have variable coefficients. Therefore, a numerical procedure, involving the use of series or special functions, is necessary to obtain solutions. The procedure is, evidently, quite cumbersome. Fortunately, for the purpose of design, it has been demonstrated by Salvadori[6] that the effect of moment gradient on the critical moment can easily be accounted for by

where

$$M_{ocr} = \text{Eq. (5.4.34)}$$

$$C_b = 1.75 + 1.05\left(\frac{M_A}{M_B}\right) + 0.3\left(\frac{M_A}{M_B}\right)^2 \leq 2.3 \tag{5.5.2}$$

in which (M_A/M_B) is the ratio of the numerically smaller to larger end moments. Its value is positive when the beam bends in double curvature and is negative when the beam bends in single curvature.

A comparison of the theoretical value of M_{cr} with that evaluated from Eqs. (5.5.1) for various values of M_A/M_B is shown in Fig. 5.14. One can see that Eq. (5.5.1) gives a conservative and quite accurate representation of the actual critical moment. It should be noted that the concept of equivalent uniform moment used here for beams is very similar to that used earlier in Chapter 3 for beam-columns. The physical meaning of C_b here is that it represents the amount of increase in the

5.5.2 Central Concentrated Load

If a simply supported beam is loaded at midspan by a concentrated force, the moment diagram is bilinear, as shown in Fig. 5.15. Here, as in the case of unequal end moments, the differential equation will contain a variable coefficient.

As an illustration, consider a simply supported I-beam subjected to a concentrated force P applied at the shear center of the middle cross section (Fig. 5.16). To derive the governing differential equation, we need to relate the externally induced moments acting on the beam at its slightly deformed (buckled) configuration to its internal resistance. The procedure is facilitated by using the two coordinate systems: a fixed coordinate (x-y-z) system and a local coordinate (x'-y'-z') system as shown in Fig. 5.16. As the beam buckles laterally, vertical reactions $P/2$ and torsional reactions $Pu_m/2$, where u_m is the lateral out-of-plane displacement of the shear center of the middle cross section, will develop at the supports. By considering a cross section at a distance z from the

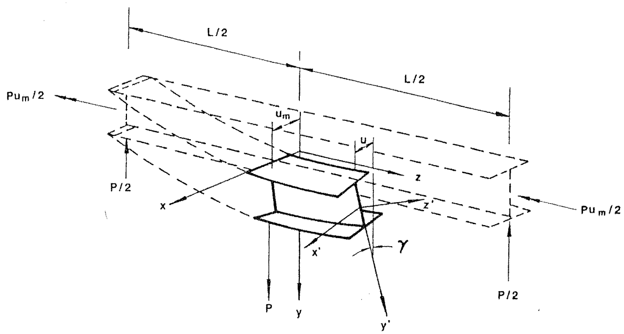
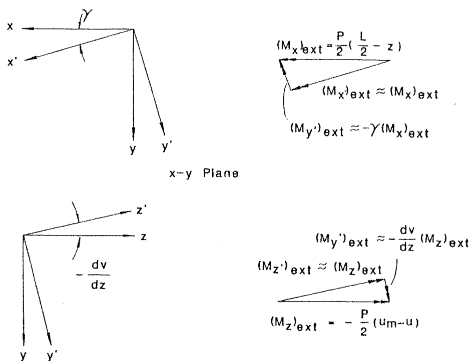


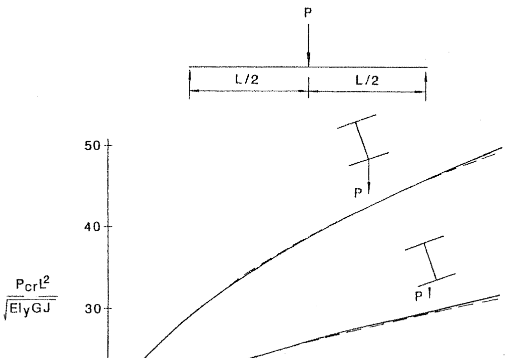
FIGURE 5.16 Lateral buckling of a simply supported I-beam loaded at midspan



Note that the second term in Eq. (5.5.6) and (5.5.7) is neglected in writing Eq. (5.5.12) and (5.5.13), because the quantities (du/dz), (dv/dz), and $(u_m - u)$ are all small. The reader should recognize that Eq. (5.5.12), which describes the in-plane bending behavior of the beam, is uncoupled with the other two equations. Therefore, it is not important in the present buckling analysis. The lateral torsional buckling behavior of the beam is described by Eq. (5.5.13) and Eq. (5.5.14). By eliminating u from Eqs. (5.5.13) and (5.5.14) and noting that $du_m/dz = 0$, we can write a simple differential equation as

$$EC_w \frac{d^4\gamma}{dz^4} - GJ\frac{d^2\gamma}{dz^2} + \frac{1}{EI_y}\left[\frac{P}{2}\left(\frac{L}{2} - z\right)\right]^2 \gamma = 0 \qquad (5.5.15)$$

This differential equation has a variable coefficient in its third term. The solution for this differential equation can be obtained[2] by the method of infinite series. The results are plotted as solid lines in Fig. 5.18. The



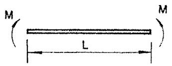
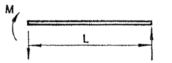
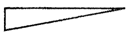
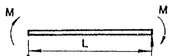
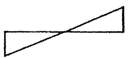
plotted as dotted lines on the same figure. It can be seen that the approximate solutions give an excellent representation of the theoretical exact solutions.

5.5.3 Other Loading Conditions

The effect of the distribution of load along the unbraced length of a simply supported beam on its elastic lateral torsional buckling strength has been investigated numerically by a number of researchers. The results are discussed in various books,[2,8,9] and papers.[10–15] For simplicity, approximate solutions in the form of Eq. (5.5.1) are often used to obtain the critical loads. The approximate solutions for some commonly encountered loading cases with the loads applied at the shear center of the cross section are summarized in Table 5.2a. By using the expressions for M_{cr} in the third column and the value of C_b in the fourth column, together with M_{ocr} given in Eq. (5.4.34), the corresponding approximate

Table 5.2 Values of C_b for Different Loading Cases (All Loads are Applied at Shear Center of the Cross Section)

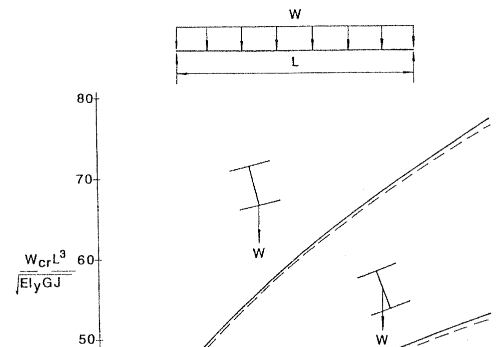
(a)

Loadings	Bending Moment Diagrams	M_{cr}	C_b
		M_{cr}	1.00
		M_{cr}	1.75
		M_{cr}	2.30

...ads Applied at Bottom Flange, Shear

A	B
1.12	$1+0.535W-0.154W^2$
$1+(\frac{L_1}{L})^2$	$1+1.636W-0.465W^2$

...nge
...r
...ge



supported beam by imagining the beam to be consisted of two cantilevers of equal length joined together at the fixed ends. Thus, the critical moment for the cantilever beam can be obtained from Eq. (5.4.34) by replacing L by $2L$

$$M_{cr} = \left(\frac{\pi}{2L}\right)(\sqrt{EI_yGJ})\sqrt{\left(1 + \frac{\pi^2 EC_w}{(2L)^2 GJ}\right)} \tag{5.6.1}$$

For other loading conditions, recourse must be made to numerical procedures to obtain solutions.[17,18] Figures 5.20 and 5.21 show the results for two load cases: cantilever beam with a concentrated load at the free end and cantilever beam with a uniform distributed load. For both of these load cases, the figures present the critical loads corresponding to

FIGURE 5.20 Critical loads of a cantilever subjected to a concentrated force at the free end

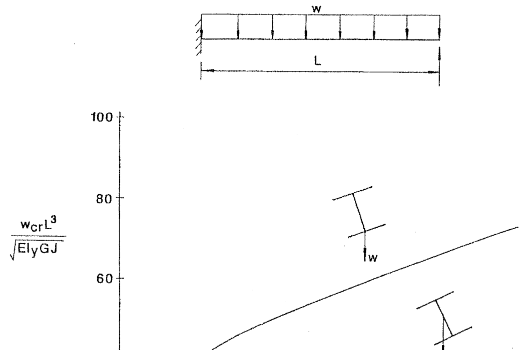
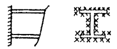
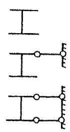
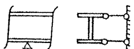
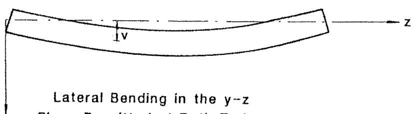


Table 5.4 Effective Length Factors for Cantilevers with Various End Conditions (Adapted from Ref. 16)

Restraint Conditions		Effective Length	
At root	At tip	Top flange loading	All other cases
		1.4L	0.8L
		1.4L	0.7L
		0.6L	0.6L
		2.5L	1.0L

Warping and Lateral Bending in the
x–z Plane Prevented at Both Ends



Lateral Bending in the y–z

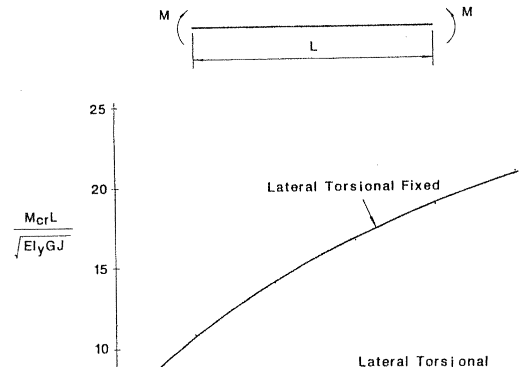


Table 5.5 Expressions of A and B for a Fixed-End Beam $[W = (\pi/L)\sqrt{EC_w/GJ}]$

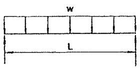
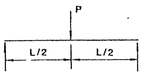
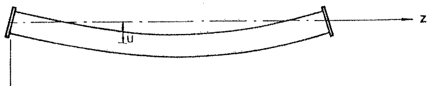
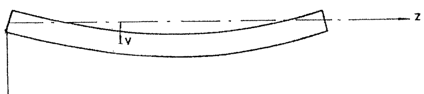
Load Case	A	B
	$1.643+1.771W-0.405W^2$	$1+0.625W-0.339W^2$
	$1.916+1.851W-0.424W^2$	$1+0.923W-0.466W^2$

Table 5.6 Effective Length Factors for Beams Under Uniform Moment with Various Boundary Conditions (Adapted from Ref. 19)

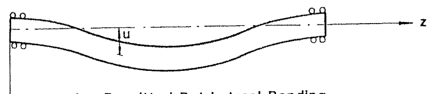
Boundary Conditions		K_b	K_t
$z = 0$	$z = L$		
$u = u'' = \gamma = \gamma'' = 0$	$u = u'' = \gamma = \gamma'' = 0$	1.000	1.000
$u = u'' = \gamma = \gamma'' = 0$	$u = u'' = \gamma = \gamma' = 0$	0.904	0.693
$u = u'' = \gamma = \gamma'' = 0$	$u = u' = \gamma = \gamma'' = 0$	0.626	1.000
$u = u'' = \gamma = \gamma'' = 0$	$u = u' = \gamma = \gamma' = 0$	0.693	0.693
$u = u'' = \gamma = \gamma' = 0$	$u = u'' = \gamma = \gamma' = 0$	0.883	0.492
$u = u' = \gamma = \gamma'' = 0$	$u = u' = \gamma = \gamma' = 0$	0.431	0.693
$u = u' = \gamma = \gamma' = 0$	$u = u' = \gamma = \gamma' = 0$	0.492	0.492
$u = u' = \gamma = \gamma'' = 0$	$u = u' = \gamma = \gamma'' = 0$	0.434	1.000
[illegible]	[illegible]	0.606	0.492



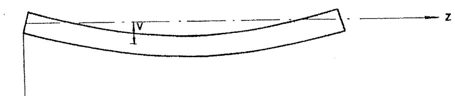
Warping Prevented But Lateral Bending in the x–z Plane Permitted at Both Ends



Lateral Bending in the y–z



Warping Permitted But Lateral Bending in the x–z Plane Prevented at Both Ends



Lateral Bending in the y–z

Table 5.8 Expressions of A and B for a Warping Permitted but Lateral Bending Prevented Beam [$W = (\pi/L)\sqrt{EC_w/GJ}$]

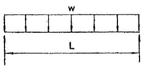
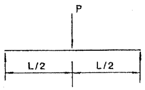
Load Case	A	B
	$1.9+0.006W-0.120W^2$	$1+0.806W-0.100W^2$
	$2.0+0.304W-0.074W^2$	$1+1.047W-0.207W^2$

Table 5.9 Values of C_1 and C_2 in Eq. (5.6.16) (Adapted from Ref. 20)

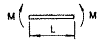
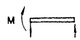
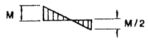
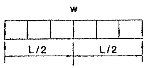
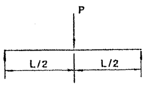
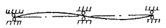
Case No.	Loading	Bending Moment Diagram	Condition of Restraint Against Rotation about Vertical Axis at Ends	Equiv. Length Factor, K	Value of Coefficients C_1^*	Value of Coefficients C_2
Beams Restrained Against Lateral Displacement at Both Ends of Span						
1.			Simple support Fixed	1.0 0.5	1.0 1.0	0 0
2.			Simple support Fixed	1.0 0.5	1.3 1.3	0 0
3.			Simple support Fixed	1.0 0.5	1.8 1.8	0 0
4.			Simple support Fixed	1.0 0.5	2.4 2.3	0 0

Table 5.10 Expressions of A and B for a Two-Span Continuous Beam with Lateral Support at Center and Restraint Equal at Both Ends $[W = (\pi/L)\sqrt{EC_w/GJ}]$

Load Case	A	B
	$2.093+3.117W-0.947W^2$	$1.073+0.044W$
	$2.95+4.070W-1.143W^2$	1



where δ_0 and θ_0 are the initial out-of-straightness and twist at the midspan of the beam, the out-of-plane bending and torsional equations [Eqs. (5.4.8) and (5.4.9)] become

$$EI_y\frac{d^2u}{dz^2} + (\gamma + \gamma_0)M_0 = 0 \tag{5.7.3}$$

$$GJ\frac{d\gamma}{dz} - \left(\frac{du}{dz} + \frac{du_0}{dz}\right)M_0 = 0 \tag{5.7.4}$$

Differentiating Eq. (5.7.4) once and substituting the result into Eq. (5.7.3), we obtain

$$EI_yGJ\frac{d^2\gamma}{dz^2} + \gamma M_0^2 = EI_yM_0\frac{d^2u_0}{dz^2} - \gamma_0M_0^2 \tag{5.7.5}$$

In view of Eqs. (5.7.1) and (5.7.2), we can write Eq. (5.7.5) as

Equation (5.7.11) can be satisfied for all values of z only if both the coefficients of the sine and cosine terms vanish. That is

$$C = \frac{-k^2\theta_0\left(\dfrac{M_{ocr}}{M_0}+1\right)}{k^2-\left(\dfrac{\pi}{L}\right)^2} \tag{5.7.12}$$

and either

$$D = 0 \tag{5.7.13}$$

or

$$k^2 = \left(\frac{\pi}{L}\right)^2 \tag{5.7.14}$$

However, if Eq. (5.7.14) is valid, then the solution reduces to Eq. (5.4.16); which is not the case to be investigated here. So, Eq. (5.7.13) must be valid. Thus the particular solution reduces to

Backsubstituting Eq. (5.7.18) into Eq. (5.7.3) or Eq. (5.7.4), and making use of Eqs. (5.7.1), (5.7.2), and (5.7.7), it can be shown that

$$u = \left[\frac{\delta_0 M_0/M_{ocr}}{1-(M_0/M_{ocr})}\right]\sin\frac{\pi z}{L} \tag{5.7.19}$$

The total twist and out-of-plane deflection are obtained by adding Eqs. (5.7.18) and (5.7.19) to Eqs. (5.7.2) and (5.7.1), respectively,

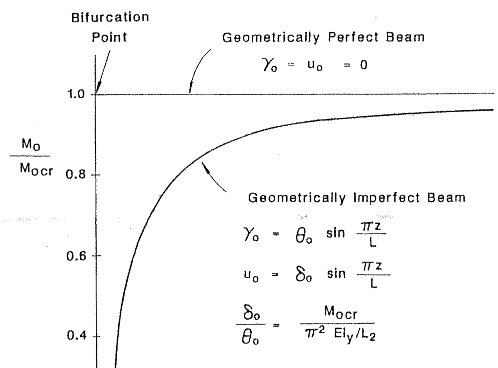
$$\gamma_{total} = \gamma_0 + \gamma = \left[\frac{1}{1-(M_0/M_{ocr})}\right]\theta_0 \sin\frac{\pi z}{L} = A_F\gamma_0 \tag{5.7.20}$$

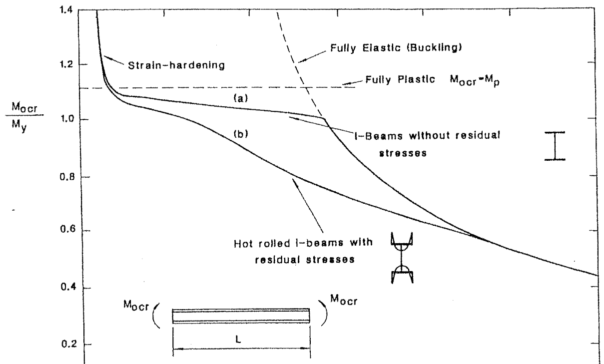
and

$$u_{total} = u_0 + u = \left[\frac{1}{1-(M_0/M_{ocr})}\right]\delta_0 \sin\frac{\pi z}{L} = A_F u_0 \tag{5.7.21}$$

where

$$A_F = \frac{1}{1-(M_0/M_{ocr})} \tag{5.7.22}$$


Bifurcation Point
Geometrically Perfect Beam
$\gamma_o = u_o = 0$
1.0
$\frac{M_o}{M_{ocr}}$
0.8
Geometrically Imperfect Beam
0.6
$\gamma_o = \theta_o \sin \frac{\pi z}{L}$
$u_o = \delta_o \sin \frac{\pi z}{L}$
$\frac{\delta_o}{\theta_o} = \frac{M_{ocr}}{\pi^2 EI_y/L_2}$
0.4


1.4
1.2
1.0
0.8
0.6
0.4
0.2
$\frac{M_{ocr}}{M_y}$
Strain-hardening
Fully Elastic (Buckling)
Fully Plastic $M_{ocr} = M_p$
(a)
(b)
I-Beams without residual stresses
Hot rolled I-beams with residual stresses
M_{ocr}
M_{ocr}
L

To obtain the inelastic buckling loads for beams with a more general loading case other than that of equal and opposite end moments, recourse must be had to numerical methods.[24,27,28] The reason for this is that for beams under general loadings, the in-plane bending moment will vary along the beam, and so the distribution of yielding will also vary from cross section to cross section. Nethercot and Trahair[29] have presented numerical solutions for some hot-rolled I-beams with a number of different loading arrangements. Some of the results are shown in Fig. 5.28, in which the nondimensional moment M_B/M_p (M_p = plastic moment capacity of the section) is plotted as a function of the modified slenderness $\sqrt{M_y/M_{ocr}}$. Note that the most severe loading case is the one with equal and opposite end moments that bends the beam in a single curvature and the least severe case is the one with equal end moment that bends the beam in a double curvature. For the former loading case, yielding occurs along the entire length of the beam, whereas for the latter loading case, yielding is usually confined to small portions at or near the

strain hardening of the material, this strain-hardening effect is usually neglected in design consideration.

5.9 DESIGN CURVES FOR STEEL BEAMS

It is quite evident from the preceding discussions that the lateral instability behavior of real beams is complex. Some of the important parameters are (1) the lateral unsupported length of the beam, (2) the cross-sectional geometry, (3) the material behavior (elastic versus inelastic), (4) the magnitude, type, and location of the applied loads, and (5) the type of lateral supports, as well as the end conditions of the beam, etc. Because of the complexity of this problem, certain simplifying assumptions are inevitable in order to make use of the analytical results for design purposes. In this section, we shall briefly discuss some aspects of the design rules as specified by the AISC Specifications.[30,31] Three different design approaches, already familiar to the reader, will be

4. The depth-thickness ratio of the web or webs must satisfy the criterion given by the following applicable equation. For $f_a/F_y \leqslant 0.16$,

$$\frac{d}{t_w} \leqslant \frac{640}{\sqrt{F_y}}\left(1 - 3.74\frac{f_a}{F_y}\right) \tag{5.9.2}$$

For $f_a/F_y > 0.16$,

$$\frac{d}{t_w} \leqslant \frac{257}{\sqrt{F_y}} \tag{5.9.3}$$

In Eqs. (5.9.2) and (5.9.3) [AISC Eqs. (1.5-4a) and (1.5-4b)],
d = the depth of the section
t_w = the thickness of the web
f_a = the applied axial stress in the section (in ksi)
F_y = the yield stress of the material (in ksi)

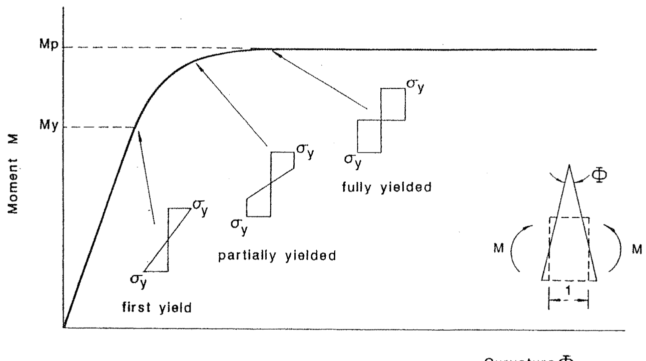
Sections that do not satisfy the requirements stated above are referred to as noncompact sections. For noncompact sections, local buckling may

The allowable bending stress F_b for the member will be discussed as follows:

ALLOWABLE STRESS F_b
The *allowable bending stress* F_b in the beam with lateral unbraced length L_b is as follows.

Compact Section with $L_b \leqslant L_c$. If the unbraced length L_b is less than or equal to L_c, the beam is considered to be adequately braced, and so lateral instability will not be a factor. For such cases, the full plastic moment of the section can be developed. Thus, the allowable bending stresses for doubly symmetric I-section (except hybrid girders and members of A514 steel) bent about the strong axis are

$$F_b = 0.66F_y \tag{5.9.6a}$$



Compact or Noncompact Section with $L_b > L_u$ Bent About the Strong Axis.

For

$$r_T\sqrt{\frac{102{,}000C_b}{F_y}} \leqslant L_b \leqslant r_T\sqrt{\frac{510{,}000C_b}{F_y}}$$

$$F_b = \text{larger of}\begin{cases}\left[\dfrac{2}{3} - \dfrac{F_y(L_b/r_T)^2}{1530\times 10^3 C_b}\right]F_y & (5.9.10a)\\[2ex] \dfrac{12{,}000C_b}{L_b d/A_f} & (5.9.10b)\end{cases}$$

but must not exceed $0.60F_y$, and for

$$L_b \geqslant r_T\sqrt{\frac{510{,}000C_b}{F_y}}$$

then

Equations (5.9.10a) and (5.9.11a) are Eqs. (1.5-6a) and (1.5-6b) in the AISC Specification, respectively, and Eqs. (5.9.10b) and (5.9.11b) correspond to Eq. (1.5–1.7) in the AISC Specification. In these equations, C_b is a parameter to account for the moment gradient in the member as defined in Eq. (5.5.2) for unequal end moments. Although values for C_b for other load conditions are available (Table 5.2a), a conservative value of unity is used for design purposes, when the bending moment at any point within an unbraced length is larger than that at both ends of the unbraced length. In such a case, the reduction in stiffness due to the yielding material usually exceeds the benefits of the less severe pattern of nonuniform moment distribution and the use of $C_b = 1.0$ is therefore more appropriate. C_b is also taken as unity for frames braced against joint translation and for cantilever beams.

Equations (5.9.10a) and (5.9.11a) were developed by assuming that the torsional resistance of the beam is negligible compared to warping resistance. The former equation represents lateral buckling of the

section. If we take

$$G = \frac{E}{2(1+\nu)} = \frac{E}{2.6}$$

$$I_y = A r_y^2, \qquad C_w = I_y h^2/4$$

$$S_x = 2A r_x^2/d, \qquad h \approx 0.95d$$

$$r_x \approx 0.41d, \qquad J \approx 0.28 A t_f^2$$

where

ν = Poisson's ratio (= 0.3 for steel)
r_x, r_y = radius of gyration about the x and y axis, respectively
d = overall depth of beam
t_f = flange thickness
h = distance between centroids of flanges

and substitute it into Eq. (5.9.13), we can write can be written as

$$F_{ocr} = \frac{0.66E}{L_b d/A_f} \tag{5.9.17}$$

To account for the moment gradient, the parameter C_b is used; thus,

$$F_{cr} = C_b F_{ocr} = \frac{0.66EC_b}{L_b d/A_f} \tag{5.9.18}$$

Finally, substitution of $E = 29{,}000$ ksi into Eq. (5.9.18) and dividing the result by a safety factor of 1.67 gives

$$F_b \approx \frac{12{,}000C_b}{L_b d/A_f} \quad \text{(in ksi)} \tag{5.9.19}$$

Equation (5.9.19) is Eq. (5.9.10b) and Eq. (5.9.11b) [AISC Eq. (1.5-7)].

CONSIDERATION OF WARPING STRENGTH ONLY

Equation (5.9.22) is valid so long as all the fibers in the column compression flange remain elastic. This, in turn, can be realized so long as the unbraced length of the beam is large. If it is small, inelastic buckling will occur and the SSRC (or CRC) parabola will again be used to present this inelastic buckling strength:

$$F_{cr} = \left[1 - \frac{F_y}{4\pi^2 E}\left(\frac{L_b}{r_T}\right)^2\right] F_y \tag{5.9.23}$$

Using the term C_b to account for moment gradient and dividing the equation by a safety factor of 1.50, we obtain

$$F_b = \left[\frac{2}{3} - \frac{F_y (L_b/r_T)^2}{1720 \times 10^3 C_b}\right] F_y \tag{5.9.24}$$

The number 1720×10^3 in the second term is replaced by 1530×10^3 to account for the fact that as L_b/r_T gets larger, the reserved strength due to strength. To assure this, the AISC Specification provides the limits on the width-to-thickness ratios of flanges and webs as well as the limits on the lateral unbraced length of the beam.

Local Buckling

LIMITS FOR WIDTH-TO-THICKNESS RATIO FOR FLANGES

F_y, ksi	$b_f/2t_f$
36	8.5
42	8.0
45	7.4
50	7.0
55	6.6
60	6.3

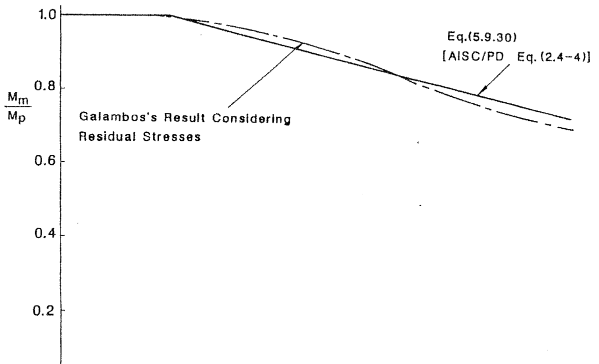
where

M/M_p = the ratio of the end moments at the ends of the unbraced segment, positive when the segment is bent in reverse curvature and negative when bent in single curvature

r_y = the radius of gyration of the member about its weak axis in inches

F_y = the yield stress of the material (in ksi)

Equations (5.9.28) and (5.9.29) are AISC Eqs. (2.9-1a) and (2.9-1b), respectively. They are applicable for members bent about their strong axis, but not applicable for members bent about their weak axis, nor are they applicable in the region of the last plastic hinge formation in the development of a failure mechanism. This is because for members bent about their weak axis, lateral instability is not a factor, and so the length



$$L_r = \frac{r_y X_1}{(F_y - F_r)} \sqrt{(1 + \sqrt{(1 + X_2 (F_y - F_r)^2)})} \tag{5.9.34}$$

$$L_b = \text{unbraced length of the member} \tag{5.9.35}$$

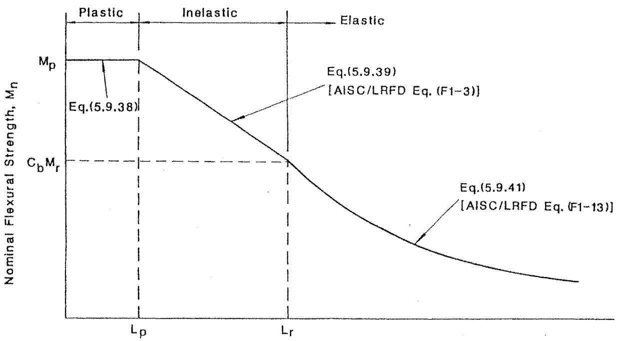
where

r_y = radius of gyration about the weak axis in inches
F_y = yield stress in ksi

$$X_1 = \frac{\pi}{S_x} \sqrt{\frac{EAGJ}{2}} \tag{5.9.36}$$

$$X_2 = \frac{4C_w}{I_y} \left(\frac{S_x}{GJ}\right)^2 \tag{5.9.37}$$

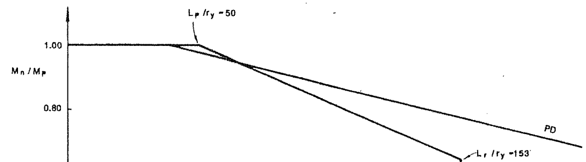
S_x = section modulus about strong axis (in in.^{3})
E = elastic modulus (in ksi)



Note that Eq. (5.9.41) is identical to the exact elastic buckling solution given by Eq. (5.5.1). In other words, the theoretical lateral torsional buckling moment of an elastic beam under equal end moments is used directly in LRFD.

Equations (5.9.39), (5.9.34), and (5.9.41) are contained in the AISC/

FIGURE 5.33 Comparison of beam curves for ASD, PD, and LRFD



LRFD Specification[31] as Eqs. (F1-3), (F1-6), and (F1-13), respectively. Figure 5.32 shows schematically the variation of the nominal moment capacity M_n with the unbraced length L_b. The LRFD beam curve is generally more liberal and much simpler than that of the ASD beam curve.

The equations given above are applicable only to compact nonhybrid I-sections in which yielding or lateral torsional instability is the limit state. The applicable expressions for other sections, including symmetric box sections, solid rectangular bars, hybrid sections, and tee and double-angle sections, are given elsewhere.[31]

Figure 5.33a,b shows a comparison of the variation of the nondimensionalized nominal moment capacity M_n/M_p as a function of the slenderness ratio L_b/r_y of a beam using the three design approaches (ASD, PD, and LRFD) discussed in the preceding sections. The beam is a W18 × 76 section with a yield stress $F_y = 36$ ksi and a compressive residual stress in the flange $F_r = 10$ ksi. The Young's modulus E is taken

present two other empirical methods to represent beam strength in the transition range.

5.10.1 Structural Stability Research Council Approach

In the Structural Stability Research Council (SSRC) approach, it is assumed that the stability behavior of beams is analogous to the stability behavior of columns, and so the lateral-torsional buckling strength of a beam is represented by the SSRC multiple-column curves. Using the Rondal–Maquoi form [Eq. (2.11.17)] with P/P_y replaced by M_n/M_p and λ_c replaced by λ_b, where $\lambda_b = \sqrt{M_p/M_{cr}}$, we have

$$\frac{M_n}{M_p} = \frac{(1 + \eta + \lambda_b^2) - \sqrt{[(1 + \eta + \lambda_b^2)^2 - 4\lambda_b^4]}}{2\lambda_b^2} \tag{5.10.1}$$

where η is defined the same as for columns in Eq. (2.11.18).

1.52 in., $J = 0.545$ in.^{4}, $Z_x = 64.0$ in.^{3}, $C_w = 1450$ in.^{6}

$$C_b = 1.0 \quad \text{(constant moment)}$$

$$M_p = Z_x F_y = 2304 \text{ in-kips}$$

$$M_{cr} = C_b M_{ocr} = M_{ocr} = \frac{\pi}{L_b}\sqrt{\left[EI_y GJ + \left(\frac{\pi^2}{L_b^2}\right)EI_y EC_w\right]}$$

$$= 2765 \text{ in-kips}$$

1. *AISC/ASD*:

$$L_c = \text{smaller of}\begin{cases} \dfrac{76 b_f}{\sqrt{F_y}} = 88.5 \text{ in.} \\ \dfrac{20{,}000}{(d/A_f)F_y} = 105 \text{ in.} \end{cases}$$

$$= 88.5 \text{ in.}$$

2. *AISC/LRFD*

$$L_p = \frac{300 r_y}{\sqrt{F_y}} = 76 \text{ in.}$$

$$L_r = \frac{r_y X_1}{F_y - F_r}\sqrt{[1 + \sqrt{[1 + X_2(F_y - F_r)^2]}]}$$

$$= 219 \text{ in.}$$

Since $L_p < L_b < L_r$, therefore, by using Eq. (5.9.39) with $M_r = S_x(F_y - F_r)$, we get

$$M_n = C_b\left[M_p - (M_p - M_r)\left(\frac{L_b - L_p}{L_r - L_p}\right)\right]$$

$$= 1544 \text{ in-kips}$$

3. *SSRC*

instability is a typical bifurcation problem. We can use the concept of neutral equilibrium to obtain the critical load. In this approach, a slightly deformed state of the beam corresponding to its buckled position is first drawn and equilibrium equations are then written with respect to this deformed configuration. The eigenvalue solution to the characteristic equation of the resulting equilibrium equation gives the *critical load* of the beam. Depending on the type and nature of the loading condition, the resulting linear differential equations may have constant or variable coefficients. For the later case, it is often necessary to resort to numerical techniques to obtain solutions.

The critical loads for beams subjected to loads that induce moment gradients can be obtained approximately by introducing the parameter C_b, as in Eq. (5.5.1). This parameter was developed on the basis of *equivalent moment concept.* Thus, the product $C_b M_{ocr}$ in Eq. (5.5.1) represents the magnitude of a pair of equal and opposite end moments that would cause lateral instability in the beam, the same as real loadings

easily be obtained as a function of the corresponding initial deformations [Eqs. (5.7.20) and (5.7.21)].

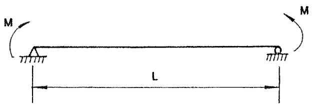
The assumption of a perfectly elastic behavior is justified only if the slenderness ratio of the beam is large. If this is not the case, material yielding will occur before buckling, with the result that only the elastic core of the cross section is effective in resisting the buckling deformation. The study of the inelastic buckling behavior of I-beams is beyond the scope of this book; it is given in detail in reference 24. Nevertheless, the readers should realize that material yielding has a detrimental effect on the buckling strength of the beam. Consequently, the buckling load of an inelastic beam usually falls significantly below the elastic buckling load. For design purposes, the inelastic buckling strength of beams with intermediate slenderness ratio is represented, therefore, by a simple curve-fitting technique. For example, the ASD uses the parabola approximation [Eq. (5.9.10a)], whereas the PD and LRFD uses a straight line approximation [Eqs. (5.9.30) and (5.9.39), respectively].

behavior of a tip-loaded cantilever beam of rectangular cross section if the load is applied at the following:

a. at the centroid of the cross section
b. at the top of the cross section
c. at the bottom of the cross section

5.3 Plot the critical moment M_{ocr} as a function of the length L for a rectangular beam subjected to a pair of equal and opposite end moments (Fig. P5.3a)

FIGURE P5.3



(a)

with the cross sections shown in Fig. P5.3b if

$$t = 1 \text{ in.} \quad (25.4 \text{ mm})$$
$$E = 29{,}000 \text{ ksi} \quad (2 \times 10^5 \text{ MN/m}^2)$$
$$G = 12{,}000 \text{ ksi} \quad (8.3 \times 10^4 \text{ MN/m}^2)$$
$$J = C\tfrac{1}{3}bt^3$$

where C is a correction factor that is a function of b/t as shown in Fig. P5.3c.
What observations can you make?

5.4 Show that Eq. (5.5.22) can be used to approximate the values for C_b given in the fourth column of Table 5.2a (Page 334).

5.5 Plot the beam curves for a simply supported W8 × 31 section using

a. LRFD approach

Determine the elastic buckling value of the concentrated load P that acts at the free end of the top flange using the relevant solution curves given in this chapter.

5.10 A simply supported steel I-beam whose properties are given below has a central concentrated load applied in such a way that lateral deflection and twist are prevented at midspan ($u_m = \gamma_m = 0$). If the beam span is 35 ft, then

a. determine the elastic buckling load M_{cr}
b. determine the nominal moment capacity M_n specified in the LRFD Specification

where

$A = 18.35\ \text{in.}^2$, $I_x = 1330\ \text{in.}^4$, $r_x = 8.54\ \text{in.}$, $S_x = 126.4\ \text{in.}^3$, $Z_x = 144.1\ \text{in.}^3$,

$I_y = 57.5\ \text{in.}^4$, $r_y = 1.77\ \text{in.}$, $r_T = 1.93\ \text{in.}$, $C_w = 5968\ \text{in.}^6$, $J = 1.97\ \text{in.}^4$,

$E = 30{,}000$ ksi, $G = 12{,}000$ ksi, $F_y = 45$ ksi

(see Fig. P5.10).

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Chapter 6

ENERGY AND NUMERICAL METHODS

stability) *approach,* must be used. In this approach, the complete load-deflection response of the member is traced from the start of loading up to the critical load. Such an analysis for the relatively simple case of *elastic frameworks* has been described in Section 4.5, Chapter 4. Rigorous load-deflection analysis for inelastic, imperfect members is rather complicated. As a consequence, recourse must be had to numerical means. We shall discuss two such numerical methods in the later part of this chapter.

In this chapter, we will present two approximate methods of stability analyses. They are the *energy method* and the *numerical method.* In the energy method, an approximate value for the critical load is determined by examining the variation and balance of energies before and after buckling in a structural system. It is valid only for structural systems that are elastic and conservative. For an inelastic system, recourse must be had to numerical methods to obtain solutions. We will present two such methods in this chapter. They are the *Newmark method* and the

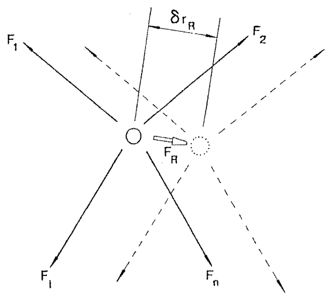
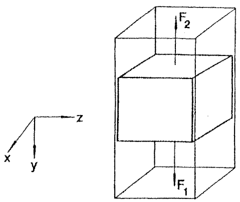
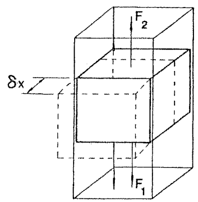


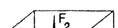
FIGURE 6.1 Particle subjected to a system of forces



(a)



(b)



mass is in equilibrium, then Eq. (6.2.3) to Eq. (6.2.5) must vanish.

$$\delta W = R_x \delta x = 0 \tag{6.2.6}$$

$$\delta W = R_z \delta z = 0 \tag{6.2.7}$$

$$\delta W = F_1 \delta y - F_2 \delta y = 0 \tag{6.2.8}$$

Since none of the arbitrary displacements (δx, δz, δy) are zero, from Eq. (6.2.6) we must have $R_x = 0$ and from Eq. (6.2.7), $R_z = 0$. Finally, from Eq. (6.2.8), we must have $F_1 = F_2$. However, the fact that $R_x = R_z = 0$ is quite obvious by inspection of Fig. 6.2a. Therefore, the only useful information we can obtain concerning the equilibrium conditions of the rigid mass is from Eq. (6.2.8), which states that if the mass is in equilibrium, the applied forces F_1 and F_2 must be equal in magnitude, opposite in direction and applied on the same straight line. Note that since Eq. (6.2.8) was obtained by introducing an arbitrary displacement δy that does not violate the condition of geometric in which q_i are the *generalized displacements* of the system and δq_i are independent variations of the generalized displacements. F_i are the *generalized forces* acting in the direction of q_i. The word *generalized* is used here to emphasize that the displacements can be either translational or rotational and that the forces can be concentrated forces or moments. If the forces are distributed, Eq. (6.2.10) still applies, provided that the summation sign is replaced by an integral sign.

For systems that are not rigid, deformation will occur when forces are applied to the systems. As a result, in addition to doing *external work*, the systems will do *internal work* when virtual displacements are introduced. Thus, the virtual work equation for the system that is not rigid, viz. a deformable system, takes the form

$$\delta W = \delta W_{\text{ext}} + \delta W_{\text{int}} = 0 \tag{6.2.11}$$

where

δW_{ext} = external virtual work

To show the validity of Eq. (6.2.11), let us consider a system that consists of a particle attached to three springs (Fig. 6.3a) subjected to a system of external forces $F_{1\,\text{ext}}, F_{2\,\text{ext}}, \ldots, F_{n\,\text{ext}}$. Under the action of this set of external forces, the particle will displace. This displacement will induce internal forces $F_{1\,\text{int}}, F_{2\,\text{int}}, F_{3\,\text{int}}$ in the springs (Fig. 6.3b). Denoting $F_{\text{R ext}}$ and $F_{\text{R int}}$ as the resultants of the set of external forces and the set of internal forces, respectively, and F_{R} as the resultant of all the forces (external and internal) acting on the particle (Fig. 6.3c), if a virtual displacement δr_{R} is introduced to the particle in its equilibrium position in the direction of F_{R} (Fig. 6.3d), the total virtual work done on the particle will be

$$\begin{aligned}
\delta W &= \delta W_{\text{ext}} + \delta W_{\text{int}} \\
&= F_{\text{R ext}}\delta r_{\text{R}} - F_{\text{R int}}\delta r_{\text{R}} \\
&= F_{\text{R}}\delta r_{\text{R}}
\end{aligned} \tag{6.2.12}$$

mass W_{mass}. In other words, we must have

$$F_{spring} = W_{mass} \tag{6.2.13}$$

Now, let us establish this equilibrium condition using the principle of virtual work. Suppose we introduce a virtual displacement δy from the equilibrium position of the system (Fig. 6.4c). The virtual work done by the external force on the mass is

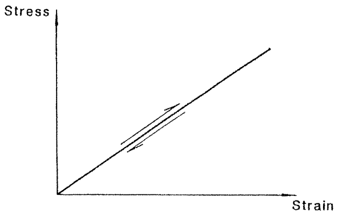
$$\delta W_{ext} = W_{mass}\delta y \tag{6.2.14}$$

and the virtual work done by the internal force on the mass is

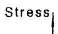
$$\delta W_{int} = -F_{spring}\delta y \tag{6.2.15}$$

The internal virtual work is negative because the direction of internal force F_{spring} is opposite to that of the virtual displacement δy.

Using Eq. (6.2.11), we have



(a) Linear Elastic System



work can be related to the variation of virtual strain energy in the system by

$$\delta W_{\text{int}} = -\delta U \tag{6.3.3}$$

If the external forces acting on the system are conservative (that is, if the work done by these forces is path independent and fully recoverable in a loading/unloading cycle), the increase in the external work done on the system will correspond to a decrease in the potential energy of the system. As a result, we can also write

$$\delta W_{\text{ext}} = -\delta V \tag{6.3.4}$$

where V is the *potential energy* of the system.

Upon substituting Eqs. (6.3.3) and (6.3.4) into Eq. (6.2.11), we obtain

$$-\delta V - \delta U = 0 \tag{6.3.5}$$

or

$$\delta(U + V) = 0 \tag{6.3.6}$$

6.4 CALCULUS OF VARIATIONS

The *calculus of variations* is concerned with the evaluation of stationary values or extremals (maximum or minimum quantities) of *functionals*. Functionals are definite integrals of function(s). The function(s) in these integrals are unknown(s) and the calculus of variations is used to determine the conditions for which these function(s) will assume a stationary value.

The calculus of variations differs from ordinary calculus in that the former deals with functionals while the latter deals with functions. Functionals are functions of functions, whereas functions are just functions of variables. In applying the calculus of variations to extremize a functional, one does not obtain the function. Instead, one obtains conditions that the function(s) must satisfy to ensure that the functional will assume a stationary value. In contrast, in applying the ordinary calculus to extremize a function, one obtains the variable(s) for which the function will assume an extremum value.

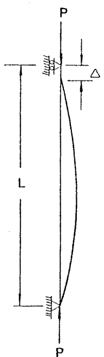
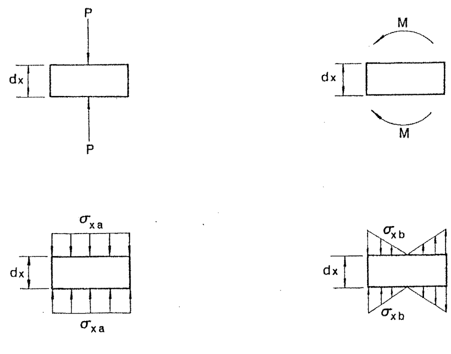


FIGURE 6.6 Hinged-hinged column



we have

$$U_a = \frac{1}{2}\int_0^L \int_A \frac{1}{E}\left(\frac{P}{A}\right)^2 dA\, dx$$

$$= \frac{1}{2}\int_0^L \frac{P^2}{EA} dx \tag{6.4.6}$$

Equation (6.4.6) can be expressed in terms of the axial strain $\varepsilon_{xa} = du/dx$, where u is the axial displacement by recognizing that

$$P = EA\varepsilon_{xa} = EA\frac{du}{dx} \tag{6.4.7}$$

therefore

$$U_a = \frac{1}{2}\int_0^L EA\left(\frac{du}{dx}\right)^2 dx \tag{6.4.8}$$

Strain Energy Due to Bending, U_b. Figure 6.8b shows the stress

Recognizing that $\int_A y^2\, dA = I$, the moment of inertia of the cross section, we can write

$$U_b = \frac{1}{2}\int_0^L \frac{M^2}{EI} dx \tag{6.4.14}$$

An alternate form for U_b in terms of the curvature d^2v/dx^2 instead of the moment M can be obtained if we substitute $M = -EI\left(\frac{d^2v}{dx^2}\right)$ into Eq. (6.4.14)

$$U_b = \frac{1}{2}\int_0^L EI\left(\frac{d^2v}{dx^2}\right)^2 dx \tag{6.4.15}$$

where v is the lateral deflection of the column.

By combining Eq. (6.4.8) with Eq. (6.4.15), the total strain energy of the column in Fig. 6.6 is

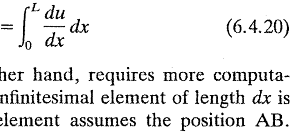


FIGURE 6.9 Bending shortening of an infinitesimal element

can be obtained by integrating the axial strain $\varepsilon_{xa} = du/dx$ over the length of the column, that is

$$\Delta_a = \int_0^L \varepsilon_{xa}\, dx = \int_0^L \frac{du}{dx}\, dx \tag{6.4.20}$$

The bending shortening, on the other hand, requires more computations. Consider Fig. 6.9, in which an infinitesimal element of length dx is shown. Before bending occurs, the element assumes the position AB. After bending, the element assumes the position AB'. The differential

Total Potential Energy of the Column

By combining Eqs. (6.4.16) and (6.4.26), the total potential energy of the column can be expressed as

$$\begin{aligned}\Pi &= U + V \\ &= \frac{1}{2}\int_0^L EA\left(\frac{du}{dx}\right)^2 dx + \frac{1}{2}\int_0^L EI\left(\frac{d^2v}{dx^2}\right)^2 dx \\ &\quad - P\int_0^L \left(\frac{du}{dx}\right) dx - \frac{P}{2}\int_0^L \left(\frac{dv}{dx}\right)^2 dx\end{aligned} \tag{6.4.27}$$

For equilibrium, the first variation of the total potential energy of the column must vanish

$$\delta\Pi = \delta(U + V) = \delta U + \delta V = 0 \tag{6.4.28}$$

since

letting

$$p=\frac{du}{dx}\quad\text{and}\quad dq=\frac{d\,\delta u}{dx}\,dx \tag{6.4.32}$$

and knowing

$$\int_0^L p\,dq = pq\,\Big|_0^L-\int_0^L q\,dp \tag{6.4.33}$$

we have

$$EA\int_0^L\left(\frac{du}{dx}\right)\left(\frac{d\,\delta u}{dx}\right)dx = EA\frac{du}{dx}\,\delta u\,\Big|_0^L - EA\int_0^L\frac{d^2u}{dx^2}\,\delta u\,dx \tag{6.4.34}$$

Similarly, for the second term

$$EI\int_0^L\left(\frac{d^2v}{dx^2}\right)\left(\frac{d^2\,\delta v}{dx^2}\right)dx$$

(6.4.38) into Eq. (6.4.36), we have

$$EI\int_0^L\left(\frac{d^2v}{dx^2}\right)\left(\frac{d^2\,\delta v}{dx^2}\right)dx = EI\left(\frac{d^2v}{dx^2}\right)\left(\frac{d\,\delta v}{dx}\right)\Big|_0^L + EI\int_0^L\frac{d^4v}{dx^4}\,\delta v\,dx \tag{6.4.39}$$

For the third term of Eq. (6.4.31)

$$-P\int_0^L\frac{d\,\delta u}{dx}\,dx$$

Using integration by parts [Eq. (6.4.33)] with

$$p=1\quad\text{and}\quad dq=\frac{d\,\delta u}{dx}\,dx \tag{6.4.40}$$

we can write

or

$$\left(EA\frac{du}{dx}-P\right)\delta u\big|_{x=L}-\left(EA\frac{du}{dx}-P\right)\delta u\big|_{x=0}-\int_0^L EA\frac{d^2u}{dx^2}\,dx\,\delta u$$
$$+\left(EI\frac{d^2v}{dx^2}\right)\left(\delta\frac{dv}{dx}\right)\Big|_{x=L}-\left(EI\frac{d^2v}{dx^2}\right)\left(\delta\frac{dv}{dx}\right)\Big|_{x=0}$$
$$+\int_0^L\left(EI\frac{d^4v}{dx^4}+P\frac{d^2v}{dx^2}\right)dx\,\delta v$$
$$=0 \tag{6.4.46}$$

Since δv and δu are arbitrary and can take on any value for the interval 0 to L, to ensure that Eq. (6.4.46) is satisfied regardless of the value of δv and δu, each and every term in Eq. (6.4.46) must vanish. Therefore, we have, from the first term, either

or

$$\delta\left(\frac{dv}{dx}\right)\Big|_{x=0}=0 \tag{6.4.55}$$

and from the sixth term

$$\int_0^L\left(EI\frac{d^4v}{dx^4}+P\frac{d^2v}{dx^2}\right)dx=0$$

or

$$EI\frac{d^4v}{dx^4}+P\frac{d^2v}{dx^2}=0 \tag{6.4.56}$$

Close scrutiny of Eq. (6.4.46) shows that the first three terms contain only the axial displacement, u, as the variable, whereas the last three terms contain only the transverse or lateral displacement, v, as the variable. Since they are uncoupled, we can treat them separately. In fact, (6.4.50) express the geometric (or kinematic) boundary conditions of the column. For the column shown in Fig. 6.6, it is obvious that the axial displacement at the end $x = L$ is not zero, therefore $\delta u \neq 0$ at $x = L$. As a result, we must have from Eq. (6.4.47)

$$EA \frac{du}{dx}\bigg|_{x=L} = P \tag{6.4.57}$$

At $x = 0$, the end is pinned, therefore we must have

$$\delta u|_{x=0} = 0 \quad \text{or} \quad u|_{x=0} = 0 \tag{6.4.58}$$

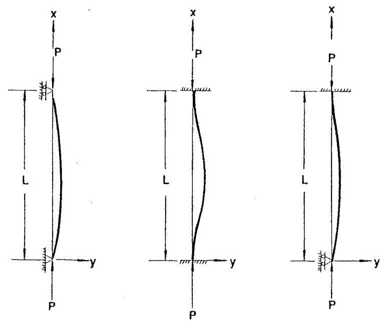
In other words, the axial force at the top end of the column is equal to P and the axial displacement at the bottom end of the column is equal to zero.

Equation (6.4.51) is simply the differential equation for an axially loaded column subjected to concentrated force at its ends.

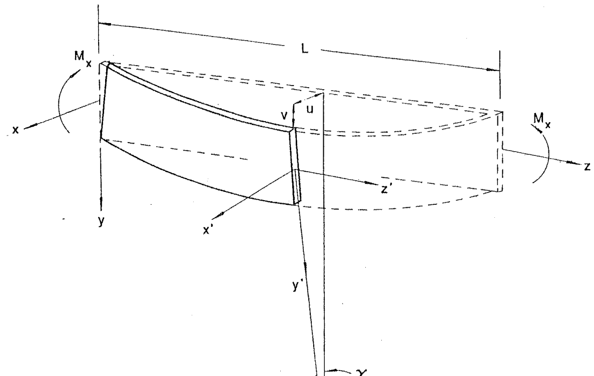
the expressions for the strain energy U and the potential energy V will not change. Although it is quite obvious that U will not change because it is independent of the boundary conditions of the member, the fact that V, which is dependent upon end conditions and loadings of the members, will not change for a fixed-fixed or a fixed-pinned column is a fact that deserves some explanation. For example, for a fixed-fixed column, the expression for V is

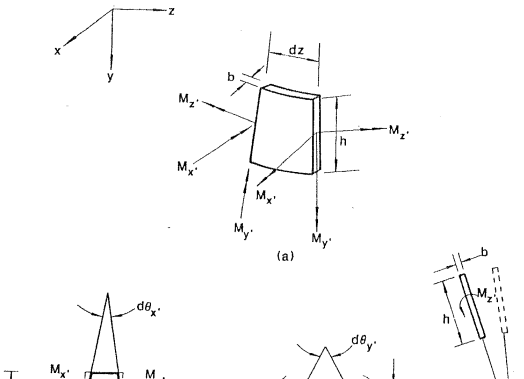
$$V = -P\int_0^L \left(\frac{du}{dx}\right) dx - \frac{P}{2}\int_0^L \left(\frac{dv}{dx}\right)^2 dx - \left(M\frac{dv}{dx}\right)\bigg|_{x=0} - \left(M\frac{dv}{dx}\right)\bigg|_{x=L} \tag{6.4.61}$$

The first two terms in Eq. (6.4.61) were explained earlier in this section: they are the work done by the axial force P as the force P travels through an axial shortening Δ_a and a bending shortening Δ_b. The third and fourth terms represent the work done as the end moment acts through an end



Differential Equation	$EIy^{IV}+Py''=0$	$EIy^{IV}+Py''=0$	$EIy^{IV}+Py''=0$





we have

$$dW_{\text{ext}} = \frac{1}{2} EI_x \left(\frac{d^2v}{dz^2}\right)^2 dz \tag{6.4.67}$$

and so

$$W_{\text{ext}} = \frac{1}{2} \int_0^L EI_x \left(\frac{d^2v}{dz^2}\right)^2 dz \tag{6.4.68}$$

In view of Eq. (6.4.63), we have

$$U_{bx'} = \frac{1}{2} \int_0^L EI_x \left(\frac{d^2v}{dz^2}\right)^2 dz \tag{6.4.69}$$

Strain Energy Due to Out-Of-Plane Bending, $U_{by'}$. The increment of external work performed as a result of out-of-plane bending is equal to

From Eq. (6.4.63), we can write

$$U_{by'} = \frac{1}{2}\int_0^L EI_y\left(\frac{d^2u}{dz^2}\right)^2 dz \tag{6.4.75}$$

Strain Energy Due to Twisting, $U_{tz'}$. The increment of external work performed as a result of twisting is equal to one-half the product of the twisting moment $M_{z'}$ and the change in the angle of twist $d\gamma$ (Fig. 6.12d)

$$dW_{ext} = \tfrac{1}{2}M_{z'}\, d\gamma \tag{6.4.76}$$

Since, from mechanics of material (see Eq. 5.2.5, page 311)

$$M_{z'} = GJ\frac{d\gamma}{dz} \tag{6.4.77}$$

where

G = shear modulus
J = torsional constant

With the strain energy expression for the beam in Fig. 6.11 developed, the next step is to develop the expression for the potential energy for the beam.

Potential Energy

The potential energy of the beam in its buckled configuration consists of two parts: the potential energy developed as a result of in-plane bending (V_1) and the potential energy developed as a result of out-of-plane bending and twisting (V_2). They are described as follows:

Potential Energy Due to In-Plane Bending, V_1. The potential energy due to in-plane bending of the beam is equal to the negative product of the applied moments M_x and the end rotations (Fig. 6.13a):

$$V_1 = -\left[M_x\left(-\frac{dv}{dz}\right)\bigg|_{z=L} + M_x\left(\frac{dv}{dz}\right)\bigg|_{z=0}\right] = M_x\left(\frac{dv}{dz}\right)\bigg|_0^L \tag{6.4.83}$$

Potential Energy Due to Out-of-Plane Bending and Twisting, V_2. The potential energy due to out-of-plane bending and twisting is equal to the negative product of the applied moments M_x and the rotation of the end cross sections α as the beam buckles

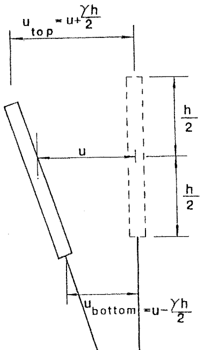
$$V_2 = -2M_x\alpha \tag{6.4.84}$$

where, as in Fig. 6.13b,

$$\alpha = \frac{1}{2}\,\frac{\Delta_{\text{top}} - \Delta_{\text{bottom}}}{h} \tag{6.4.85}$$

In Eq. (6.4.85), Δ_{top} and Δ_{bottom} are the bending shortening of the top and bottom fibers, respectively, of the cross section, while h is the height of the cross section.

These bending shortenings can be expressed as



For equilibrium, the first variation of the total potential energy of the beam must vanish, that is

$$\begin{aligned}\delta\Pi = &\int_0^L EI_x\left(\frac{d^2v}{dz^2}\right)\delta\left(\frac{d^2v}{dz^2}\right)dz + \int_0^L EI_y\left(\frac{d^2u}{dz^2}\right)\delta\left(\frac{d^2u}{dz^2}\right)dz\\ &+\int_0^L GJ\left(\frac{d\gamma}{dz}\right)\delta\left(\frac{d\gamma}{dz}\right)dz + M_x\delta\left(\frac{dv}{dz}\right)\Big|_0^L\\ &-M_x\int_0^L\left(\frac{du}{dz}\right)\delta\left(\frac{d\gamma}{dz}\right)dz - M_x\int_0^L\left(\frac{d\gamma}{dz}\right)\delta\left(\frac{du}{dz}\right)dz\\ =&\,0\end{aligned} \tag{6.4.95}$$

or

$$\begin{aligned}&\int_0^L EI_x\left(\frac{d^2v}{dz^2}\right)\left(\frac{d^2\,\delta v}{dz^2}\right)dz + \int_0^L EI_y\left(\frac{d^2u}{dz^2}\right)\left(\frac{d^2\,\delta u}{dz^2}\right)dz\\ &+\int_0^L GJ\left(\frac{d\gamma}{dz}\right)\left(\frac{d\,\delta\gamma}{dz}\right)dz + M_x\,\delta\left(\frac{dv}{dz}\right)\Big|_0^L\end{aligned}$$

or

$$EI_x\frac{d^4v}{dz^4}=0 \tag{6.4.100}$$

and for the third term

$$EI_y\frac{d^2u}{dz^2}=0 \quad \text{at} \quad z=0 \quad \text{and} \quad L \tag{6.4.101}$$

or

$$\delta\left(\frac{du}{dz}\right)=0 \quad \text{at} \quad z=0 \quad \text{and} \quad L$$

and for the fourth term

$$\int_0^L\left(EI_y\frac{d^4u}{dz^4}+M_x\frac{d^2\gamma}{dz^2}\right)dz=0$$

or

expressed in Eqs. (6.4.101) to (6.4.103) apply. Again, because the beam is simply supported, it is obvious that $\delta(du/dz) \neq 0$ and $z = 0$ and L. Therefore, Eq. (6.4.101) will apply. Equation (6.4.101) expresses the natural boundary conditions of the beam that state that the moment about the y axis vanishes at the simply supported ends.

Equations (6.4.102) and (6.4.103), which correspond, respectively, to Eqs. (5.4.8) and (5.4.9), express the lateral torsional buckling behavior of the beam. By eliminating the variable u, these two equations can be combined to give

$$\frac{d^4\gamma}{dz^4} + k^2 \frac{d^2\gamma}{dz^2} = 0 \qquad (6.4.105)$$

where

$$k^2 = \frac{M_x^2}{GJEI_y} \qquad (6.4.106)$$

Equation (6.4.105) corresponds to Eq. (5.4.11) in Chapter 5. Equation

applied to establish the governing differential equations of a system. However, the solutions to the differential equations are not obtained in this extremization process. For practical purposes, it is also necessary to obtain the critical loads from these governing differential equations. Theoretically, the exact elastic critical loads can be solved from these equations using the method of eigenvalue analysis described in Chapter 2. For many practical problems, however, the exact solutions to the characteristic equations of the differential equations are often difficult, if not impossible, to obtain, and recourse must be had to approximate or numerical methods to obtain solutions. In the following, we shall present a simple method to obtain an approximate solution of the critical load. This method is called the *Rayleigh–Ritz method*. In this method, an assumed displacement function satisfying the *geometric boundary conditions* of the system is used in the expression for the total potential energy function Π. By using such an assumed displacement function, a structural system with an infinite degree of freedom is now reduced to a system of configuration of the member is identified if

$$\delta(\bar{U} + \bar{V}) = 0 \tag{6.5.4}$$

Since $\bar{U}$ and $\bar{V}$ are functions of a, we can write Eq. (6.5.4) as

$$\frac{\partial(\bar{U} + \bar{V})}{\partial a}\,\delta a = 0 \tag{6.5.5}$$

or, because δa is arbitrary, we must have

$$\frac{\partial(\bar{U} + \bar{V})}{\partial a} = 0 \tag{6.5.6}$$

The value of P satisfying Eq. (6.5.6) is the critical load of the member.
To obtain a better result, a series of arbitrary functions ϕ_i's can be used as the assumed deflected shape of the member. That is, we can let

Example 6.1. Critical Load of a Cantilever Column Loaded at the Tip
Figure 6.15a shows a cantilever column loaded by an axial force P at the tip. The exact elastic buckling load, $P_{cr} = \pi^2 EI/4L^2 = 2.47EI/L^2$, has been obtained in Chapter 2. We now want to obtain an approximate solution of the critical load using the Rayleigh–Ritz method.

SOLUTION: Assume the deflection shape of the cantilever column to be

$$\bar{v} = ax^2 \tag{6.5.12}$$

Since $\bar{v}(0) = \bar{v}'(0) = 0$, the assumed deflection function satisfies the geometric boundary conditions of the member.
The total potential energy of the member is

$$\begin{aligned}\Pi &= U + V \\ &= \frac{1}{2}\int_0^L EI\left(\frac{d^2v}{dx^2}\right)^2 dx - \frac{P}{2}\int_0^L \left(\frac{dv}{dx}\right)^2 dx \end{aligned} \tag{6.5.13}$$

or

$$4aEIL - 4PaL^3/3 = 0$$

from which we have

$$P_{cr} = 3\frac{EI}{L^2} \tag{6.5.15}$$

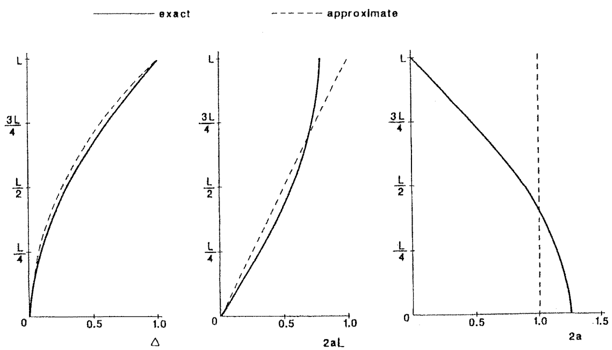
Comparing with the exact elastic critical load of $2.47EI/L^2$, we see that the approximate critical load is 21% in error.

The approximate solution can be improved if we use Eq. (6.4.14) rather than Eq. (6.4.15) for the strain energy of the column. The moment expression for the column can be written as

$$M = P(\Delta - v) \tag{6.5.16}$$

where Δ is the lateral tip deflection of the column (Fig. 6.15b).

From Eq. (6.5.12), we know



the total potential energy can be written as

$$\bar{\Pi} = \bar{U} + \bar{V}$$
$$= \frac{1}{2}\int_0^L EI\left(\frac{d^2\bar{v}}{dx^2}\right)^2 dx - \frac{P}{2}\int_0^L \left(\frac{d\bar{v}}{dx}\right)^2 dx$$
$$= \frac{1}{2}\int_0^L EI(2a_1 + 6a_2x)^2\, dx$$
$$- \frac{P}{2}\int_0^L (2a_1x + 3a_2x^2)^2\, dx$$
$$= 2EIL(a_1^2 + 3a_1a_2L + 3a_2^2L^2)$$
$$- \frac{PL^3}{30}(20a_1^2 + 45a_1a_2L + 27a_2^2L^2) \tag{6.5.22}$$

Note that the total potential energy function is now a function of two

The smallest positive root of this equation, called the *characteristic equation*, is $\lambda = 2.49$ and from Eq. (6.5.25), we obtain

$$P_{cr} = 2.49\frac{EI}{L^2} \tag{6.5.31}$$

which differs from the exact solution by only 0.81%. The solution can be further improved by using $U = \int_0^L (M^2/EI)\, dx$ in Eq. (6.5.22). This is not attempted here.

Example 6.2. Critical Load of a Cantilever Column Loaded at the Tip and Midheight

In the preceding example, the Rayleigh–Ritz method has been used to determine an approximate critical load of a relatively simple problem whose exact critical load is well known. In the following we shall consider a more difficult or rather more cumbersome example problem.

for the column shown in Fig. 6.17 in which loads are present at both the tip and at midheight.

Using Eq. (6.4.15), the strain energy of the column corresponding to the assumed deflection shape is

$$\begin{aligned}\bar{U} &= \frac{1}{2}\int_0^L EI\left(\frac{d^2\bar{v}}{dx^2}\right)^2 dx \\ &= \frac{1}{2}\int_0^L EI\left[a\left(\frac{\pi}{2L}\right)^2 \cos\frac{\pi x}{2L}\right]^2 dx \\ &= \frac{EIa^2\pi^4}{64L^3} \end{aligned} \tag{6.5.33}$$

The potential energy is

the other hand, when we use more terms for the assumed deflected shape of the member, we are mathematically introducing more degrees of freedom to the member. As a result, the stiffening effect of the constraint brought by the assumed shape is reduced.

To obtain a reasonable critical load for a problem in the Rayleigh–Ritz method, the deflection functions chosen for any problem must be consistent with the geometric constraints of the problem. That is, the displacement functions used must satisfy the geometric boundary conditions of the problem. Furthermore, if the assumed displacement function consists of more than one term [Eq. (6.5.7)], it is advisable to use orthogonal functions for the ϕ_i's whenever possible. Functions ϕ_a and ϕ_b are said to be orthogonal if

$$\int_0^L \phi_a\phi_b\, dx = 0 \quad \text{for} \quad a \neq b \tag{6.5.36}$$

The merit of utilizing the orthogonality property of the functions will

SOLUTION: Let the assumed deflection shape of the bar at buckling be approximated by

$$\bar{v} = \sum_{i=1}^{n} a_i \phi_i = \sum_{i=1}^{n} a_i \sin\left(\frac{i\pi x}{L}\right) \tag{6.5.37}$$

The strain energy stored in the bar at buckling is then

$$\begin{aligned}
\bar{U} &= \frac{1}{2}\int_0^L EI\left(\frac{d^2\bar{v}}{dx^2}\right)^2 dx \\
&= \frac{1}{2}\int_0^L EI\left[\sum_{i=1}^{n} \frac{-a_i i^2 \pi^2}{L^2} \sin\frac{i\pi x}{L}\right]^2 dx \\
&= \frac{EI\pi^4}{2L^4}\int_0^L \left(\sum_{i=1}^{n} a_i i^2 \sin\frac{i\pi x}{L}\right)\left(\sum_{i=1}^{n} a_i i^2 \sin\frac{i\pi x}{L}\right) dx
\end{aligned} \tag{6.5.38}$$

Equation (6.5.38) can be simplified if we use the *orthogonality* property of the sine function. It can easily be shown that

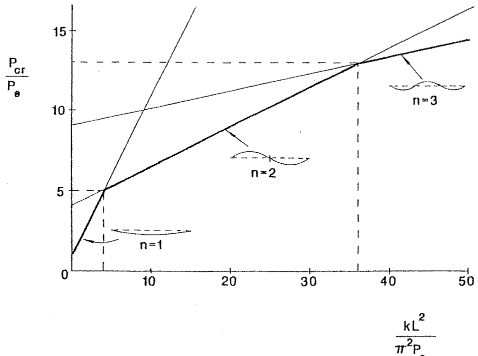
and displacement are the same. The second term represents the potential energy change due to the foundation's resistance to buckling of the bar. It is positive because the corresponding external work done is negative as the directions of the resistance force from the foundation and the lateral displacement of the bar are opposite.

To simplify Eqs. (6.5.41) and (6.5.24), we again use the orthogonality property of the terms in the assumed displacement function. For cosine function, it can easily be shown that

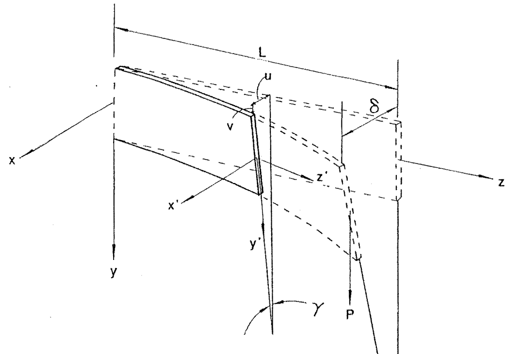
$$\int_0^L \cos ax \cos bx \, dx = 0 \quad \text{for} \quad a \neq b \tag{6.5.42}$$

and in view of Eq. (6.5.39), Eq. (6.5.41) can be simplified to

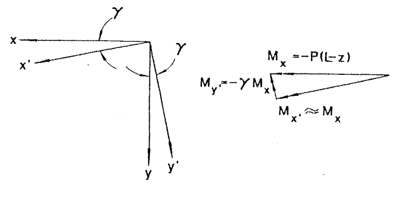
$$\bar{V} = -\frac{P\pi^2}{2L^2}\int_0^L \left(\sum_{i=1}^{n} a_i^2 i^2 \cos^2\frac{i\pi x}{L}\right) dx$$



$\frac{P_{cr}}{P_e}$
15
10
5
0
10
20
30
40
50
n=1
n=2
n=3
$\frac{kL^2}{\pi^2 P_e}$

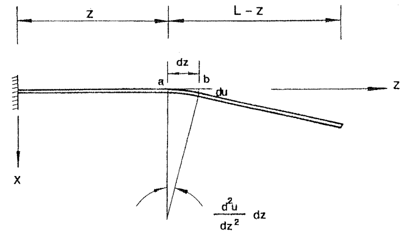


L
u
v
δ
x
z
z'
x'
y'
y
P
γ

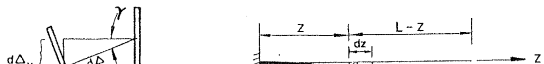

γ
x
x'
γ
M_x = -P(L-z)
M_y ≈ -γM_x
M_x' ≈ M_x
y
y'

x-y plane


dv/dz
z
z'
dv/dz
M_z = P(δ - u)


z
L - z
dz
a
b
du
z
x
d²u/dz² dz

(a)


γ
z
L - z
dz
z

member due to the applied force P is

$$V = -P\,\Delta_v = -P\int_0^L \gamma(L-z)\frac{d^2u}{dz^2}\,dz \tag{6.5.54}$$

In view of Eq. (6.5.49), we can write Eq. (6.5.54) as

$$V = -\frac{P^2}{EI_y}\int_0^L \gamma^2(L-z)^2\,dz \tag{6.5.55}$$

Combining Eqs. (6.5.50) and (6.5.55), the total potential energy of the member at buckling is

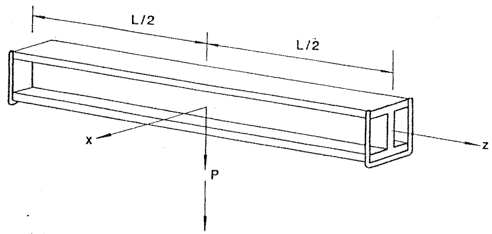
$$\begin{aligned}\Pi = U + V = &\frac{1}{2}\frac{P^2}{EI_y}\int_0^L \gamma^2(L-z)^2\,dz + \frac{1}{2}\int_0^L GJ\left(\frac{d\gamma}{dz}\right)^2 dz\\ &-\frac{P^2}{EI_y}\int_0^L \gamma^2(L-z)^2\,dz\end{aligned}$$

Integrating, we have

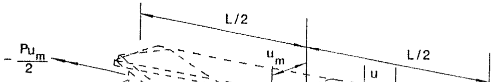
$$\begin{aligned}\bar{\Pi} = \bar{U} + \bar{V} &= -\frac{1}{2}\frac{P^2}{EI_y}\left(\frac{8\gamma_L^2L^3}{105}\right) + \frac{1}{2}GJ\left(\frac{4}{3}\frac{\gamma_L^2}{L}\right)\\ &= -\frac{4P^2\gamma_L^2L^3}{105EI_y} + \frac{2GJ\gamma_L^2}{3L}\end{aligned} \tag{6.5.60}$$

For equilibrium, we must have

$$\begin{aligned}\delta\bar{\Pi} = \delta(\bar{U} + \bar{V}) &= \frac{d(\bar{U} + \bar{V})}{d\gamma_L}\,\delta\gamma_L\\ &= \left(-\frac{8P^2\gamma_L L^3}{105EI_y} + \frac{4GJ\gamma_L}{3L}\right)\delta\gamma_L = 0\end{aligned} \tag{6.5.61}$$



(a)



term represents the strain energy due to warping restraint torsion. As indicated in Chapter 5, for all cross sections except circular sections or cross sections that are made up of thin-walled elements for which all elements intersect at a common point, the cross sections will warp. that is, plane sections will not remain plane as the member is twisted. As a result of this warping deformation, warping restraint torsion will be developed in the cross section that is not allowed to warp freely. For the case of an I-section, the strain energy stored in the member as a result of warping restraint torsion is due primarily to the coupled bending energy of the two flanges. The bending energy of the flanges is equal to

$$U_f = 2\left[\frac{1}{2}\int_0^L EI_f\left(\frac{d^2u_f}{dz^2}\right)^2 dz\right] \tag{6.5.64}$$

in which I_f is the moment of inertia of one flange about the weak axis of the cross section. The variable x is indicated in Fig. 6.24. The factor two

Substituting Eq. (6.5.65) into Eq. (6.5.64), we obtain

$$U_{\mathrm{f}} = \int_0^L EI_{\mathrm{f}}\left(\frac{h^2}{4}\right)\left(\frac{d^2\gamma}{dz^2}\right)^2 dz$$

$$= \frac{1}{2}\int_0^L EC_{\mathrm{w}}\left(\frac{d^2\gamma}{dz^2}\right)^2 dz \qquad (6.5.66)$$

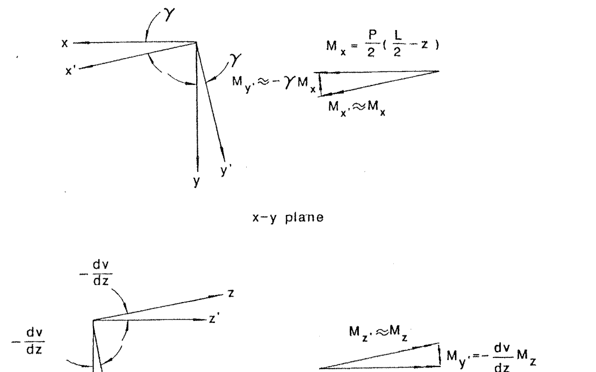
where

$$C_{\mathrm{w}} = \frac{I_{\mathrm{f}}h^2}{2} \qquad (6.5.67)$$

is the warping constant for the section.

Note that Eq. (6.5.66), which is the third term in Eq. (6.5.63), is not present in the previous case of narrow rectangular section [see Eq. (6.5.46)] because warping restraint torsion is negligible for such cross sections.

Because of symmetry, the strain energy expression of Eq. (6.5.63) can



loaded at the end by a force of $P/2$. The minus sign for the term d^2u/dz^2 accounts for the fact that the quantity d^2u/dz^2 is negative.

Upon substituting Eq. (6.5.71) into Eq. (6.5.73), the potential energy of the member at buckling can be written as

$$V = -P\,\Delta_v = -\int_0^{L/2} \frac{P^2\gamma^2}{2EI_y}\left(\frac{L}{2} - z\right)^2 dz \tag{6.5.74}$$

Combining Eqs. (6.5.72) and (6.5.74), the total potential energy for the member at buckling is

$$\Pi = U + V = \int_0^{L/2} \frac{\gamma^2 P^2}{4EI_y}\left(\frac{L}{2} - z\right)^2 dz + \int_0^{L/2} GJ\left(\frac{d\gamma}{dz}\right)^2 dz$$
$$+ \int_0^{L/2} EC_w\left(\frac{d^2\gamma}{dz^2}\right)^2 dz - \int_0^{L/2} \frac{P^2\gamma^2}{2EI_y}\left(\frac{L}{2} - z\right)^2 dz$$

By integrating, Eq. (6.5.78) can be simplified to

$$\bar{\Pi} = \bar{U} + \bar{V} = -\frac{P^2}{4EI_y}\gamma_m^2 L^3\left(\frac{1}{48} + \frac{1}{8\pi^2}\right)$$
$$+ GJ\gamma_m^2\left(\frac{\pi}{L}\right)^2\left(\frac{L}{4}\right) + EC_w\gamma_m^2\left(\frac{\pi}{L}\right)^4\left(\frac{L}{4}\right) \tag{6.5.79}$$

For equilibrium, the first variation of the total potential energy must vanish

$$\delta\bar{\Pi} = \delta(\bar{U} + \bar{V}) = \frac{d(\bar{U} + \bar{V})}{d\gamma_m}\,\delta\gamma_m$$
$$= \Big[-\frac{P^2}{2EI_y}\gamma_m L^3\left(\frac{1}{48} + \frac{1}{8\pi^2}\right)$$

6.6 GALERKIN'S METHOD

In the preceding section, the Rayleigh–Ritz method has been applied to obtain approximate values of critical loads. A somewhat similar technique that uses approximate deflected shapes to obtain approximate critical loads is a result of work done by B. G. Galerkin.[6] Unlike the Rayleigh–Ritz method, in which an expression for the total potential energy is required, Galerkin's method requires that the differential equation of equilibrium be known.

For illustration purposes, it has been shown in Section 6.4 that the total potential energy of a hinged-hinged column will assume a stationary value if the following condition is satisfied

$$\left(EI\frac{d^2v}{dx^2}\right)\left(\delta\frac{dv}{dx}\right)\bigg|_{x=L}-\left(EI\frac{d^2v}{dx^2}\right)\left(\delta\frac{dv}{dx}\right)\bigg|_{x=0}$$
$$+\int_0^L\left(EI\frac{d^4v}{dx^4}+P\frac{d^2v}{dx^2}\right)dx\ \delta v=0 \tag{6.6.1}$$

and recognize that

$$\delta\bar{v}=\delta\left(\sum_{i=1}^{n}a_i\phi_i\right)=\sum_{i=1}^{n}\phi_i\delta a_i \tag{6.6.5}$$

Thus, using Eqs. (6.6.4) and (6.6.5), Eq. (6.6.3) can be written as

$$\int_0^L Q(\bar{v})\sum_{i=1}^{n}\phi_i\delta a_i\,dx=0 \tag{6.6.6}$$

Since all the ϕ_i's are independent of one another, the only way that Eq. (6.6.6) can be satisfied is when each and every term vanishes individually, that is

$$\int_0^L Q(\bar{v})\phi_i\delta a_i\,dx=0,\qquad i=1,2,\ldots,n \tag{6.6.7}$$

Since δa_i are arbitrary, we must have

Thus, by redefining the operator Q as

$$Q = \frac{d^4}{dz^4} + k^2 \frac{d^2}{dz^2} \tag{6.6.11}$$

Eq. (6.6.8) can be used to obtain approximate value for the critical moment M_{cr}.

The following examples illustrate the use of Galerkin's method to perform approximate buckling analyses.

Example 6.6. Buckling Load of a Hinged-Fixed Column
In this example, the approximate buckling load of a hinged-fixed column as shown in Fig. 6.26a will be determined using Galerkin's method.

SOLUTION: The first step is to assume a deflection function for the buckled configuration of the column that satisfies both the kinematic and natural

column

$$\bar{y} = a(L^3x - 3Lx^3 + 2x^4) \tag{6.6.13}$$

it can easily be shown that the conditions expressed in Eq. (6.6.12) are satisfied.

Substituting the assumed deflection curve of Eq. (6.6.13) into Eq. (6.6.8) with $n = 1$, $\bar{v} = \bar{y}$, $\phi = L^3x - 3Lx^3 + 2x^4$ and

$$Q = EI\frac{d^4}{dx^4} + P\frac{d^2}{dx^2}$$

gives

$$\int_0^L a[48EI + P(-18Lx + 24x^2)][xL^3 - 3Lx^3 + 2x^4]\,dx = 0 \tag{6.6.14}$$

By integrating, Eq. (6.6.14) can be reduced to

The first two conditions are kinematic boundary conditions that state that the angle of twist at the ends are zero. The last two conditions are natural boundary conditions that state that the cross sections at the ends are free to warp.

By using the polynomial solution of the deflected shape of a simply supported beam under uniform distributed load

$$\bar{\gamma} = a(L^3z - 2Lz^3 + z^4) \tag{6.6.19}$$

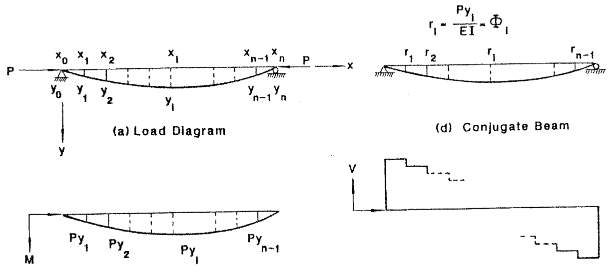
to approximate the angle of twist γ, it can easily be shown that the conditions expressed in Eq. (6.6.18) are satisfied.

Substituting the assumed function for the angle of twist [Eq. (6.6.19)] into Eq. (6.6.8) with $n = 1$, $\bar{v} = \bar{\gamma}$, $\phi = L^3z - 2Lz^3 + z^4$, and $Q = (d^4/dz^4) + k^2(d^2/dz^2)$ gives

$$\int_0^L a[24 + k^2(-12Lz + 12z^2)][L^3z - 2Lz^3 + z^4]\,dz = 0 \tag{6.6.20}$$

than the exact ones. For more complicated problems for which the true deflection curves are not known, the use of Galerkin's method in conjunction with an approximate deflection curve will furnish an *upper bound value* for the critical load.

In comparing the Rayleigh–Ritz method with Galerkin's method, two important differences can be identified. First, the Rayleigh–Ritz method deals with the total potential energy of the system, whereas Galerkin's method deals with the differential equation of equilibrium of the system. Second, the assumed deflection curve used in the Rayleigh–Ritz method need satisfy only the kinematic boundary conditions, whereas the assumed deflection curve used in Galerkin's method must satisfy both the geometric and natural boundary conditions. The nature of the problem will decide which method will be more appropriate to use. For example, if it is easier to evaluate the expression for the total potential energy function and to find an approximate deflection curve that satisfies only the kinematic boundary conditions, the Rayleigh–Ritz method will be

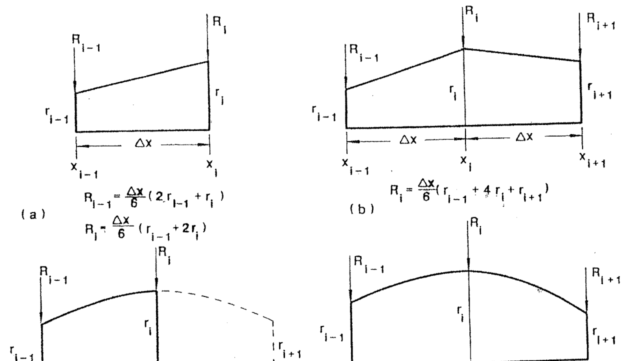


(a) Load Diagram

(b) Moment Diagram

(d) Conjugate Beam

(e) Conjugate Beam Shear



Having replaced the actual distributed loading, namely, the M/EI diagram, by equivalent concentrated loads at the stations of the conjugate beam (Fig. 6.27d), the slope of the real member can then be obtained as the shear force of the conjugate beam (Fig. 6.27e) and the deflection as the moment of the conjugate beam (Fig. 6.27f). Since equivalent concentrated loads are used, rather than the actual distributed load, the slopes evaluated represent only the *average* slopes of two adjacent stations.

The deflections thus obtained contain P as an unknown (see the example problem). Thus, by equating these deflections to the assumed deflections at each station, numerical values for P can be calculated. The lowest value of P obtained this way will represent a lower bound to the critical load, and the largest value of P will represent an upper bound to the critical load. Consequently, by using Newmark's method, both the upper and lower bound solutions to the actual critical load are obtained. This represents an obvious advantage over the Rayleigh–Ritz or

from

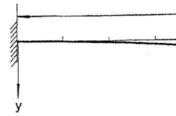
$$M_9 = P(y_{10} - y_9) = P(100 - 84.4)\frac{\delta}{100} = 15.6\frac{P\delta}{100}$$

and the moment at Station 4 is obtained from

$$\begin{aligned} M_4 &= P(y_{10} - y_4) + P(y_5 - y_4) \\ &= P(100 - 19.1)\frac{\delta}{100} + P(29.3 - 19.1)\frac{\delta}{100} \\ &= 91.1\frac{P\delta}{100} \end{aligned}$$

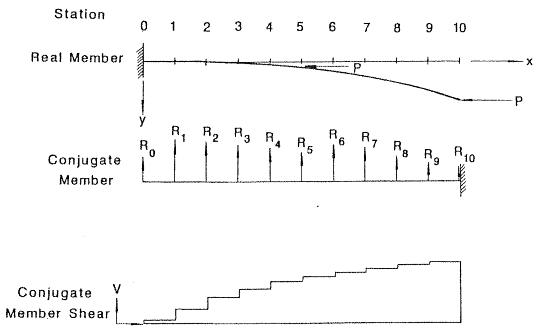
The curvature Φ can be obtained from the moment by using the elastic moment-curvature relationship given in Eq. (6.7.1). The values of the curvatures are then used in conjunction with Fig. 6.28 to calculate the

Table 6.1 Critical Load by Newmark's Method



Station	0		1		2		3	
Cycle 1								
$y_{assumed}$	0		1.23		4.89		10.9	
$\dot{M}$	129.3		127		120		108	
Φ	129.3		127		120		108	
R	6.44		12.7		12.0		10.8	
θ		6.44		19.1		31.1		4
$y_{calculated}$	0		0.64		2.55		5.66	
Ratio	/		1.91		1.92		1.93	

Station	0		1		2		3	
Cycle 2								
$y_{assumed}$	0		1.33		5.26		11.7	
M	131		128		120		108	
Φ	131		128		120		108	
R	6.53		12.8		12.0		10.8	
θ		6.53		19.3		31.3		42
$y_{calculated}$	0		0.653		2.58		5.71	
Ratio	/		2.04		2.04		2.05	



be obtained by numerically integrating the curvature and slope.

$$\theta_i = \sum_{k=0}^{i} \left(\frac{\Delta^2 y}{\Delta x^2}\right)_k \Delta x = -\sum_{k=0}^{i} \Phi_k \,\Delta x, \qquad i = 0, 1, 2, \ldots, n \quad (6.7.3)$$

$$y_i = \sum_{k=0}^{i} \left(\frac{\Delta y}{\Delta x}\right)_k \Delta x = \sum_{k=0}^{i} \theta_k \,\Delta x, \qquad i = 0, 1, 2, \ldots, n \quad (6.7.4)$$

In these formulas, Δx is the interval between the stations. The minus sign in Eq. (6.7.3) indicates that the curvature Φ is related to d^2y/dx^2 negatively as a result of the manner the coordinate axes are set up. In writing Eq. (6.7.4), it is tacitly assumed that the initial slope θ_0 is equal to zero. If θ_0 is not zero, the deflections calculated from Eq. (6.7.4) must be adjusted. The adjustment can be made by realizing that if θ_0 is not equal to zero, the deflections calculated in Eq. (6.7.4) will differ from the corrected deflections by a proportionate amount equal to $(i/n)y_n$ (Fig. 6.30). Thus the corrected deflections can be written as

FIGURE 6.31 Equivalent concentrated force for constant curvature

means that the curvature Φ_i is assumed constant over the i station, as shown in Fig. 6.31, whereas by using the conjugate beam method, one can assume a linear or parabolic variation of curvature across the station (Fig. 6.28).

Example 6.9. Inelastic Beam-Column Analysis by Newmark's Method
Figure 6.32 shows a simply supported member of rectangular cross section subjected to two end moments M_0 and $0.5M_0$ and an axial force P equal to $0.5P_y$, where P_y is the yield load of the member. Now we want to find the maximum end moment M_0 that will cause failure of the member.

SOLUTION: Before proceeding with the analysis, it is convenient to determine the following quantities from the given cross-sectional and material properties (Fig. 6.32)

$$P_y = A\sigma_y = bd\sigma_y$$

$$M_y = S\sigma_y = \frac{bd^2}{6}\sigma_y$$

Thus $M_y = (d/6)P_y = (0.06L/6)P_y(P_yL/100)$

The moment-curvature-thrust relationship used is from Eqs. (3.9.30a–c) with $p = P/P_y = 0.5$

$$m = \phi, \qquad \phi \leqslant 0.5$$

$$m = 1.5 - \sqrt{\frac{1}{2\phi}} \qquad 0.5 \leqslant \phi \leqslant 2.0$$

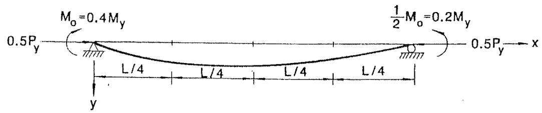
$$m = 1.125 - \frac{1}{2\phi^2} \qquad \phi \geqslant 2.0$$

from which

$$\phi = m, \qquad m \leqslant 0.5$$

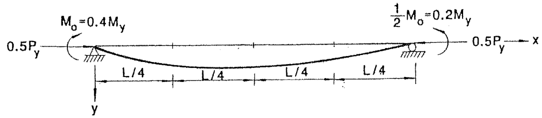
$$\phi = \frac{1}{2(1.5 - m)^2} \qquad 0.5 \leqslant m \leqslant 1.0$$

Table 6.2a Determination of Equilibrium Configuration by Newmark's Method ($M_0 = 0.4M_y$)



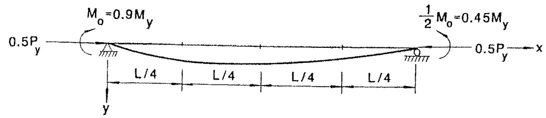
Station	0	1	2	3	4	Common Factor
Primary Moment M_0	0.4	0.35	0.30	0.25	0.2	M_y

Table 6.2a Determination of Equilibrium Configuration by Newmark's Method ($M_0 = 0.4M_y$) (*continued*)



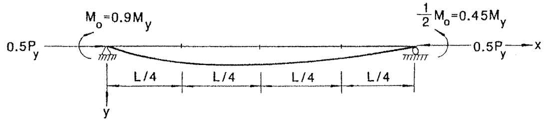
Station	0	1	2	3	4	Common Factor
Cycle 2						
y	0	0.0012	0.0015	0.001	0	L

Table 6.2b Determination of Equilibrium Configuration by Newmark's Method ($M_0 = 0.90M_y$)



Station	0	1	2	3	4	Common Factor
Primary Moment M_0	0.90	0.788	0.675	0.563	0.45	M_y

Table 6.2b Determination of Equilibrium Configuration by Newmark's Method ($M_0 = 0.90M_y$) (*continued*)



Station	0	1	2	3	4	Common Factor
Cycle 2						
y_{assumed}	0	0.0036	0.0044	0.0029	0	L

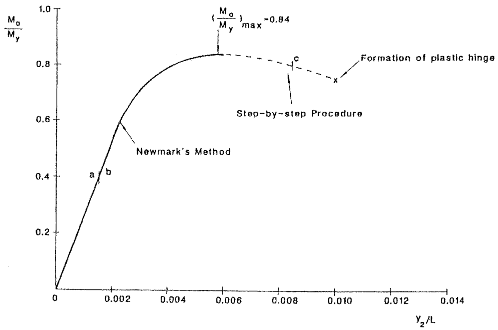


FIGURE 6.33 Load-deformation behavior of an elastic beam-column

descending branch of the curve, that is, the load-deflection behavior of the member beyond the peak point, Newmark's method cannot be used directly. Instead, another numerical procedure known as the *numerical integration procedure* should be used. This procedure is discussed in the following section.

6.8 NUMERICAL INTEGRATION PROCEDURE

In the step-by-step numerical integration procedure, the member is divided into segments, and, just as in Newmark's method, the moment-curvature-thrust relationship must be known or calculated in advance. However, unlike Newmark's method, in which an assumed deflection is assigned at each and every station, only the deflection of the first station is specified in the step-by-step numerical integration procedure. The deflections at subsequent stations are calculated from station to station in

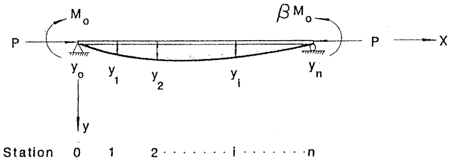
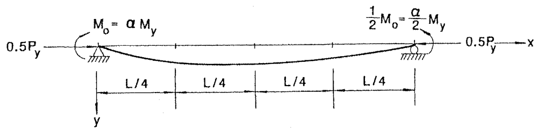


FIGURE 6.34 Step-by-step procedure

Equation (6.8.1) is the second-order central difference equation. Δx is the length of the segment.

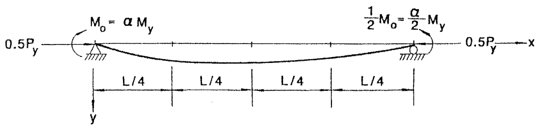
6. Knowing y_2, the secondary moment at Station 2 can be obtained by multiplying y_2 by P.
7. The total moment is obtained by adding the secondary moment to the

Table 6.3a Determination of Equilibrium Configuration by the Step-by-Step Numerical Integration Procedure ($y_1 = 0.0012L$)



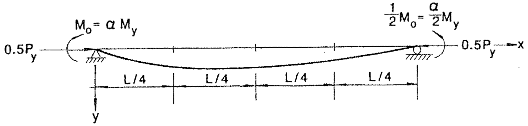
Station	0	1	2	3	4	Common Factor
Cycle 1						
Assumed Primary						

Table 6.3a *(continued)*



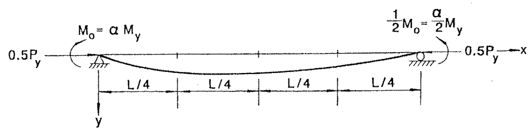
Station	0	1	2	3	4	Common Factor
Cycle 2						
Assumed Primary Moment	0.40	0.35	0.30	0.25	0.20	M_y

Table 6.3b Determination of Equilibrium Configuration by the Step-by-Step Numerical Integration Procedure ($y_1 = 0.007L$)



Station	0	1	2	3	4	Common Factor
Cycle 1						
Assumed Primary						

Table 6.3b *(continued)*



Station	0	1	2	3	4	Common Factor
Cycle 2						
Assumed Primary Moment M_0	0.804	0.704	0.603	0.503	0.402	M_y

the assumed deflection is the true deflection of the member, in which case the energy method will lead to the true buckling load of the problem. In general, the approximate critical loads represent upper bound solutions to the true critical loads.

An alternative procedure that furnishes both an upper and a lower bound to the true critical load was also presented: Newmark's method. Here the member is divided into segments and numerical values for the deflections at the stations are assumed. Through an iteration process, the true values of the deflections at the stations can be calculated. This procedure is extremely useful to evaluate the critical loads of members with a variable moment of inertia. In addition, Newmark's method is also applicable to the analysis of *inelastic* members so long as the moment-curvature-thrust relationship is known or available.

Another numerical solution scheme, presented in this chapter, that is suitable for inelastic analysis is the step-by-step numerical integration procedure. In this procedure, the numerical calculation proceeds from

6.2 Determine the approximate elastic critical load for the spring-supported column shown in Fig. P6.2 by the Rayleigh–Ritz method. k_s is the spring stiffness.

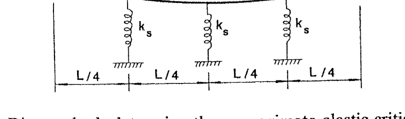


FIGURE P6.2

6.3 Using the Rayleigh–Ritz method, determine the approximate elastic critical load for the column with variable *EI* shown in Fig. P6.3.

6.5 Determine an approximate value for the elastic critical moment of the fixed-ended beam shown in Fig. P6.5 using Galerkin's method. The cross section of the beam is rectangular.

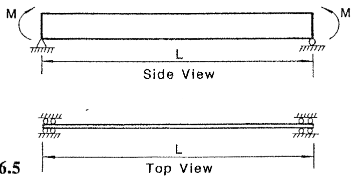


FIGURE P6.5

6.6 Using Newmark's method, determine the approximate elastic critical loads of the stepped columns shown in Fig. P6.6a–b. Establish bounds to the

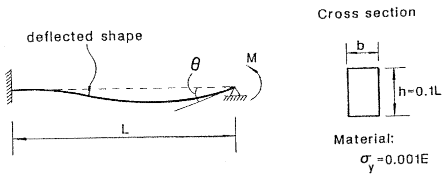


FIGURE P6.7

and for

$$m > 1,$$

$$m = \tfrac{3}{2}\left(1 - \frac{1}{3\phi^2}\right)$$

where $m = M/M_y$, $\phi = \Phi/\Phi_y$, in which M_y is the yield moment and Φ_y is the curvature that corresponds to first yield of the cross section. What is the

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ANSWERS TO SOME SELECTED PROBLEMS

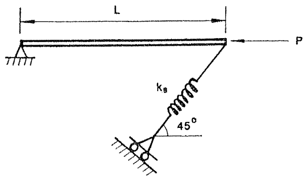

FIGURE P1.3

1.4 $P = k_s L[\cos(\alpha - \theta) - \cos\alpha]\tan(\alpha - \theta)$

$P_{cr} = 0$

$\theta < \alpha - \cos^{-1}(\cos\alpha)^{1/3}$, stable

or

b. $n = 1, \quad P = P_{cr} = \dfrac{\pi^2 EI}{4L^2}$

$n = 3, \quad P = P_{cr} = \dfrac{9\pi^2 EI}{4L^2}$

$$\frac{P_{cr(n=1)}}{P_{cr(n=3)}} = \frac{1}{9}$$

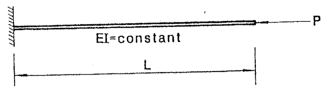

FIGURE P2.1b

2.2 $P_{cr} = \dfrac{\pi^2 EB^4}{8L^2} = 1.234\dfrac{EB^4}{L^2}$

2.11 **a.** Use W10 × 30 ASD
b. Use W10 × 30 PD
c. Use W10 × 26 LRFD

2.12 $E_r = \dfrac{2EE_t}{E + E_t}$

2.13 $P_{cr} = \dfrac{3.1EI_0}{L^2}$

2.14 $P_{cr} = \dfrac{0.1245EI}{h^2}$

CHAPTER 3 BEAM-COLUMNS

$Q \sin k(L - a)$

$$\bar{M}_{max} = \left(\frac{w}{k^2}\right)$$

$$\times \left\{ \frac{\sqrt{\left[\dfrac{k^4L^4}{4} + 2k^2L^2 + 2 - (2 + k^2L^2)(kL \sin kL + \cos kL)\right]}}{\sin kL - kL \cos kL} - 1 \right\}$$

$$M_{max} = \text{the larger of } (M_B, \bar{M}_{max}), \qquad M_0 = \frac{wL^2}{8}$$

$$M_B \approx \frac{1 - 0.4\dfrac{P}{P_{ek}}}{1 - \dfrac{P}{P_{ek}}}\left(\frac{wL^2}{8}\right)$$

b.

3.5 b. For Fig. P3.5b

$$y=\left(\frac{Q}{EIk^3}\right)\left[\frac{\cos\left(\frac{kL}{6}\right)}{\cos\left(\frac{kL}{2}\right)}\right]\sin kx-\left(\frac{Q}{EIk^2}\right)x \quad \text{for} \quad 0\leqslant x\leqslant\frac{L}{3}$$

$$y=\left(\frac{Q}{EIk^3}\right)\sin\left(\frac{kL}{3}\right)\left[\tan\frac{kL}{2}\sin kx+\cos kx\right]$$

$$-\frac{QL}{3EIk^2} \quad \text{for} \quad \frac{L}{3}\leqslant x\leqslant\frac{2}{3}L$$

$$y=-\left(\frac{Q}{EIk^3}\right)\left[\frac{\left(\sin\frac{kL}{3}+\sin\frac{2kL}{3}\right)}{\tan kL}\right]\sin kx$$

$$\left[Q\left(\sin\frac{kL}{3}+\sin\frac{2kL}{3}\right)\right.$$

d. For Fig. P3.5d

$$y=\left(\frac{Q}{EIk^3}\right)\left[\frac{\sin\frac{kL}{3}}{\sin kL}\right]\sin kx-\frac{Qx}{3EIk^2}+\left(\frac{wx}{EIk^2L}\right)\left(\frac{x^2}{2}-\frac{7L^2}{54}\right)$$

$$+\left(\frac{3w}{EIk^4L\sin kL}\right)\left(\frac{L}{3}\sin kx\cos\frac{2kL}{3}\right.$$

$$\left.+\frac{1}{k}\sin kx\sin\frac{2kL}{3}-x\sin kL\right) \quad \text{for} \quad 0\leqslant x\leqslant\frac{L}{3}$$

$$y=\left(\frac{Q}{EIk^3}\right)\left[\frac{\sin\frac{kL}{3}}{\sin kL}\right]\sin kx-\frac{Qx}{3EIk^2}-\frac{wL(L-x)}{27EIk^2}$$

$$\left(\frac{3w}{\quad}\right)\left(\frac{\sin k(L-x)}{\quad}\right)\left[\frac{1}{\quad}\quad kL\quad L\quad kL\right]$$

$\bar{x} = 0.637L$

$M_{max} = 1.3316M_B$

3.7

$$M_{FA} = M_{FB} = \frac{P\Delta(1 - \cos kL)}{kL \sin kL + 2\cos kL - 2}$$

$$= \frac{P\Delta}{kL \cot \frac{kL}{2} - 2}$$

$$k^2 = \frac{P}{EI}$$

3.13

$$M_u = \frac{PL}{1 - \frac{16P}{P_e}}$$

c. $a = 1, \quad P_{cr} = \dfrac{\pi^2 EI}{(0.58L)^2}$

4.4 $P_p = \dfrac{3.2M_p}{L}$

4.5 **a.** $P_{cr} = \dfrac{\pi^2 EI}{(0.555L)^2}$

$K = 0.555$

b. $G_T = 0.25$

$G_B = 0$

$K = 0.555$

4.6 **a.** Nonsway

5.2 **a**, **b**, and **c**

$$\frac{d^4\gamma}{dz^4} - \frac{GJ}{EC_w}\frac{d^2\gamma}{dz^2} - \frac{P^2(L-z)^2}{EC_wEI_y}\gamma = 0$$

5.3 $M_{ocr} = \dfrac{\pi}{L}\left(\dfrac{bt^3}{6}\right)\sqrt{29 \times 12 \times 10^6 c}$

$= 67.8\dfrac{bt^3}{L}\sqrt{C}\,k\text{-}ft.$

Solution 1, $2t \times t$

$$M_{ocr1} = \frac{110.379}{L}$$

Solution 2, $4t \times t$

$$M_{ocr2} = \frac{247.633}{L}$$

Solution 3, $8t \times t$

CHAPTER 6 ENERGY AND NUMERICAL METHODS

6.1 **a.** $\bar{v} = a \sin\dfrac{\pi x}{L}$

$$P_{cr} = \frac{2}{3}\frac{\pi^2 EI}{L^2} = \frac{\pi^2 EI}{(1.2247L)^2}$$

b. $\bar{v} = a(L^3x - 3Lx^2 + 2x^4)$

$$P_{cr} = \frac{23.344\pi^2 EI}{L^2} = \frac{\pi^2 EI}{(0.207L)^2}$$

c. $\bar{v} = a\left(1 - \cos\dfrac{\pi x}{2L}\right)$

$$(qL)_{cr} = 0.7992\frac{\pi^2 EI}{L^2}.$$

b. $\bar{y} = a \sin \dfrac{\pi x}{L}$

$$P_{cr} = \frac{14.8EI_0}{L^2}$$

6.5 $\bar{y} = a\left(1 - \cos \dfrac{2\pi z}{L}\right)$

$$M_{cr} = \frac{2\pi}{L}\sqrt{GJEI_y}$$

6.6 **a.** Assume the deflection shape

$$y = \delta\left(1 - \cos \frac{\pi x}{2L}\right)$$

and divide the member into 10 segments.

6.8 Divide the member into 4 segments

$$\frac{M}{M_y} = 0.4, \quad y_{\text{midspan}} = 0.00114L$$

$$\frac{M}{M_y} = 0.8, \quad y_{\text{midspan}} = 0.00372L$$

b. $M_n = 6484.5$ k-in.

INDEX

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